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Beta-product dependent Pitman–Yor processes for Bayesian inference[☆]

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ABSTRACT

Multiple time series data may exhibit clustering over time and the clustering effect may change across different series. This paper is motivated by the Bayesian non-parametric modelling of the dependence between clustering effects in multiple time series analysis. We follow a Dirichlet process mixture approach and define a new class of multivariate dependent Pitman–Yor processes (DPY). The proposed DPY are represented in terms of vectors of stick-breaking processes which determine dependent clustering structures in the time series. We follow a hierarchical specification of the DPY base measure to account for various degrees of information pooling across the series. We discuss some theoretical properties of the DPY and use them to define Bayesian non-parametric repeated measurement and vector autoregressive models. We provide efficient Monte Carlo Markov Chain algorithms for posterior computation of the proposed models and illustrate the effectiveness of the method with a simulation study and an application to the United States and the European Union business cycle.

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1. Introduction

This paper focuses on vectors of dependent Pitman–Yor (PY) processes for Bayesian inference on repeated measurements and

panel data models. The PY process of parameters $\alpha \in [0, 1)$, $\theta > -\alpha$ and base measure H_0 , in short $PY(\alpha, \theta, H_0)$, is defined by

$$G = \sum_{k \geq 1} W_k \delta_{\tilde{\psi}_k}, \quad (1)$$

where δ_x is a point mass at x , $(\tilde{\psi}_k)_k$ is a sequence of independent random variables (atoms) with common distribution H_0 , and the weights W_k 's are defined by the stick-breaking construction:

$$W_k = S_k \prod_{j < k} (1 - S_j), \quad (2)$$

S_k being independent random variables with beta distribution of parameters $(1 - \alpha, \theta + \alpha k)$. Such a process has been introduced in Pitman and Yor (1997) and it can be seen as a generalization of the (one parameter) Dirichlet process (DP) defined in Ferguson (1973). If $\alpha = 0$ the Pitman–Yor process corresponds to a DP and (1) reduces to the well-known (Sethuraman, 1994) representation.

The Dirichlet process and the PY process are usually employed as a prior for a mixing distribution, resulting in the DP mixture

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model, see [Lo \(1984\)](#). More specifically, one defines a random density

$$f(y) = \int \mathcal{K}(y|\psi)G(d\psi) = \sum_{k \geq 1} W_k \mathcal{K}(y|\tilde{\psi}_k), \quad (3)$$

where $\mathcal{K}$ is a suitable density kernel. Due to the availability of simple and efficient methods for posterior computation ([Escobar and West, 1995](#)), DP mixture models are now routinely implemented and used in many fields. A recent account of Bayesian non-parametric inference can be found in [Hjort et al. \(2010\)](#).

Although DP mixture models are extremely flexible, in many real applications data arise under different conditions and hence assuming a DP mixture model can be too restrictive. For example, using covariates, data may be divided into different units. In this case, one would like to consider different densities for different units instead of a single common density for all the units. For this reason, mixtures driven by vectors of random probability measures could be used as an alternative to the DP mixture model. After the seminal papers of [MacEachern \(1999, 2001\)](#), the problem of modelling a finite number of dependent densities, allowing information pooling across units, has become an active area of research. For the interested reader, a brief account of the state of the art is given at the beginning of Section 2.

The first contribution of this paper is to introduce a new class of dependent Pitman–Yor processes. We start from the general definition of dependent stick-breaking processes of [MacEachern \(1999\)](#) and use the multivariate beta distributions of [Nadarajah and Kotz \(2005\)](#) for the stick-breaking weights to define the beta-product dependent Pitman–Yor (DPY) process, and the beta-product dependent-Dirichlet (DD) process as a special case. Also, we present a simple and efficient method for posterior computation that constitutes a novel extension of the slice sampling algorithm for DP mixture models introduced in [Walker \(2007\)](#) and [Kalli et al. \(2011\)](#). Recently, [Hatjispyros et al. \(2011\)](#) introduce a different vector of dependent DP mixture and propose a suitable multidimensional slice sampling algorithm, which does not apply to our model.

Another contribution of the paper is the definition of two Bayesian non-parametric models for groups of time series, based on our beta-product DPY processes. The first model is an infinite mixture Gaussian model for repeated measurements which accounts for different clustering structures in the series and for dependence between the series-specific clustering effects. The model is a novel extension of the finite mixture models for repeated measurements (e.g., see [Frühwirth-Schnatter \(2006\)](#) for a review) and of the infinite mixture models for panel data (see [Hirano \(2002\)](#)).

Finally, we contribute to the literature on Bayesian vector autoregressive models (VAR) (e.g., see [Sims and Zha \(1998\)](#)) and introduce Bayesian non-parametric VAR and panel VAR models. In time series analysis DP and dependent DP have been employed in different ways. [Hirano \(2002\)](#) proposed a nonparametric autoregressive panel data model based on the DP prior, [Rodriguez and ter Horst \(2008\)](#) used a dependent DP to define an infinite mixture of time series models, [Taddy and Kottas \(2009\)](#) propose a Markov-switching finite mixture of independent Dirichlet process mixtures, [Jensen and Maheu \(2010\)](#) consider Dirichlet process mixture of stochastic volatility models, [Griffin \(2011\)](#) propose a continuous-time non-parametric model for the volatility, [Griffin and Steel \(2011\)](#) propose a time-varying stick-breaking process to capture time-variations in the clustering structure of a set of time series. In this paper, we focus on Bayesian VAR models for time series from different countries, i.e. a multi-country panel VAR (e.g., see [Chib and Greenberg \(1995\)](#) and [Canova and Ciccarelli \(2004\)](#)) and propose a non-parametric panel VAR model based

on our DPY process prior. Our model accounts for shifts in the intercept, autoregressive coefficients and covariance matrix of the country-specific equations and for cross-country dependence through a hierarchical specification of the base measure.

The structure of the paper is as follows. Section 2 introduces vectors of dependent stick-breaking processes for prior modelling in repeated measurements and vector autoregressive models. Section 3 introduces beta-product dependent Pitman–Yor processes and studies their properties. Section 4 discusses the hierarchical specification of the base measure. Section 5 proposes a Monte Carlo Markov Chain (MCMC) algorithm for approximate inference for DPY mixture models. Section 6 provides some applications to both simulated and real data. Specifically, we consider a joint analysis of two well studied business cycles of the international economic system: the United States (US) and the European Union (EU) cycle.

2. Dependent stick-breaking processes and infinite mixture models

Some of the first developments of vectors of dependent random distributions are in [Cifarelli and Regazzini \(1978\)](#). More recently, [MacEachern \(1999, 2001\)](#) introduced the dependent DP, incorporating dependence on covariates through both the atoms and the weights, although these papers mainly consider dependent atoms. In particular, the random variables $\tilde{\psi}_k$'s in the Sethuraman representation (1)–(3) are replaced with stochastic processes $\tilde{\psi}_{zk}$, z being a set of covariates. Following this line, [De Iorio et al. \(2004\)](#) propose an ANOVA-type dependence for the law of the atoms, while [Gelfand et al. \(2004\)](#) introduce a spatial dependence structure. Later, [Griffin and Steel \(2006\)](#) defined a class of DPs with both dependent atoms and weights. Many alternative constructions that incorporate dependence in the weights have been proposed, see, for instance, [Duan et al. \(2007\)](#); [Dunson et al. \(2008\)](#); [Rodriguez et al. \(2010\)](#).

Other approaches to the definition of dependent vectors of random measures rely upon suitable convex combinations of independent DPs (e.g., [Müller et al. \(2004\)](#); [Hatjispyros et al. \(2011\)](#); [Kolossiaty et al. \(2013\)](#)), hierarchical structures of stick-breakings (e.g., [Teh et al. \(2006\)](#)), normalization of dependent completely random measures (e.g., [Nipoti et al. \(in press\)](#); [Leisen and Lijoi \(2011\)](#)) or suitable multivariate extensions of the Polya tree prior (e.g., [Trippa et al. \(2011\)](#)).

2.1. Vectors of stick-breaking processes

Following the general definition of dependent stick-breaking processes, proposed in [MacEachern \(1999, 2001\)](#), we let

$$G_i(\cdot) = \sum_{k \geq 1} W_{ik} \delta_{\tilde{\psi}_{ik}}(\cdot) \quad i = 1, \dots, r, \quad (4)$$

where the weights $W_k = (W_{1k}, \dots, W_{rk})$ and the atoms $\tilde{\psi}_k = (\tilde{\psi}_{1k}, \dots, \tilde{\psi}_{rk})$ satisfy the following hypotheses:

- $(\tilde{\psi}_k)_k$ and $(W_k)_k$ are stochastically independent;
- $(\tilde{\psi}_k)_k$ is an i.i.d. sequence of random elements taking values in a product space Ψ^r with common probability distribution G_0 ;
- the weights W_{ik} s are determined via the stick-breaking construction

$$W_{ik} = S_{ik} \prod_{j < k} (1 - S_{ij}) \quad i = 1, \dots, r,$$

with $\prod_{j < 1} (1 - S_{ij}) = 1$, where $S_k = (S_{1k}, \dots, S_{rk})$ are stochastically independent random vectors taking values in $[0, 1]^r$ such that $\sum_{k \geq 1} W_{ik} = 1$ a.s. for every i .

2.2. Infinite mixture models

Modelling based on mixture distributions is nowadays applied in many areas, especially in economics and finance (see Frühwirth-Schnatter (2006) for a review). Stick-breaking processes can be used to extend finite mixture models to infinite mixture models, which can capture many specific properties of time series, such as multimodality, skewness, excess of kurtosis and presence of outliers (e.g., see Griffin (2011) and Griffin and Steel (2011)). Our interest in dependent stick-breaking processes is related to their use in building dependent infinite mixtures models, which allows for information pooling across time series. More specifically, we develop non-parametric repeated measurements and panel (or longitudinal) dynamic models. Both models are widely used in economics for joint inference on the parameters of the panel units, e.g. firms or countries, when data are collected on several occasions or observed on a regular basis.

2.2.1. Repeated measurements

Consider a set of samples, taking values in an observation space $\mathbb{Y}$, divided in r sub-samples (units or groups of observations), that is:

$$Y_{it} \quad i = 1, \dots, r, \quad t = 1, \dots, T_i$$

where Y_{it} is the t -th observation within unit i . In repeated measurements models (see Frühwirth-Schnatter (2006), Ch. 8 and references therein), i may correspond to a space label or predictor, and the observations of a unit i are independent with the same density f_i . In assessing a prior for $(f_1, f_2, \dots, f_r)$, one aims at borrowing information across units and at assuming for each unit an infinite mixture model. To do this, one can first introduce a density kernel $\mathcal{K} : \mathbb{Y} \times \Psi \rightarrow \mathbb{R}^+$, and then define

$$f_i(y) = \int \mathcal{K}(y|\psi) G_i(d\psi) \quad i = 1, \dots, r, \quad (5)$$

where $(G_1, \dots, G_r)$ is a vector of dependent stick-breaking processes.

The dependence between the random probability measures affects the dependence structure underlying the densities $f_1, \dots, f_r$, which can be represented as infinite mixtures

$$f_i(y) = \sum_{k \geq 1} W_{ik} \mathcal{K}(y|\tilde{\psi}_{ik}) \quad i = 1, \dots, r. \quad (6)$$

In this paper we focus on one of the most widely used kernel: the Gaussian kernel of parameter $\psi = (\mu, \sigma^2)$, i.e.

$$\mathcal{K}(y|\psi) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y - \mu)^2 \right\}. \quad (7)$$

Other kinds of kernels, such as binomial or multinomial, can also be applied within our framework for count, binary, and categorical data modelling (e.g., see Frühwirth-Schnatter (2006)).

2.2.2. Panel data models

A Vector Autoregressive (VAR) structure allows for handling the dynamic properties of multiple time series and is now a standard tool for structural analysis and forecasting in macroeconomics (Sims and Zha, 1998). Panel VAR models are used when different time series are collected from different units (e.g., countries). Such models can capture complex relationships between countries because they do not impose restrictions on the parameters, but this requires a large number of parameters and may lead to over-parameterization. The resulting potential over-fitting problem calls for the use of a Bayesian approach to inference, which, through the use of the prior distributions, allows for prior constraints on the VAR parameters. See, e.g., Litterman (1986),

and Sims and Zha (1998) for Bayesian VAR, Chib and Greenberg (1995) for Bayesian Seemingly Unrelated Regression and Canova and Ciccarelli (2004) for panel Bayesian VAR.

In a panel VAR model of the order p (VAR(p)), the subset of equations associated with the i th unit of the panel is:

$$Y_{it} = \mu + \sum_{j=1}^r \sum_{l=1}^p \Phi_{jl} Y_{jt-l} + \varepsilon_{it}, \quad (8)$$

for $i = 1, \dots, r$ and $t = 1, \dots, T$, where $Y_{it} = (Y_{i1,t}, \dots, Y_{im,t})'$, $\mu = (\mu_1, \dots, \mu_m)'$ and Φ_{jl} , $j = 1, \dots, r$, $l = 1, \dots, p$, is a sequence of m -dimensional square matrices. Finally, we assume $\varepsilon_{it} = (\varepsilon_{i1,t}, \dots, \varepsilon_{im,t})'$ follows a Gaussian distribution $\mathcal{N}_m(0, \Sigma)$ with mean 0 and covariance matrix Σ . Other distributions, such as heavy-tailed or long-tailed distributions, can be considered, but we leave this extension for future research. We assume that ε_{it} and ε_{jt} are independent for every $i \neq j$ and $t \neq s$. As it is common in Bayesian VAR, the dependence between units is modelled through the dependence between the random parameters of the different unit-specific subsets equations. In this section, for the sake of simplicity we dropped the dependence of the parameters μ , Φ_{jl} and Σ on the unit index i . We will discuss later on in the paper possible assumptions on the parameters of the different units and focus now on the infinite mixture representation of the unit-specific equations. First we write (8) in a stacked regression form

$$Y_{it} = (I_m \otimes X'_t) \phi + \varepsilon_{it}, \quad (9)$$

where $X_t = (1, Y'_{1t-1}, \dots, Y'_{rt-1}, \dots, Y'_{1t-p}, \dots, Y'_{rt-p})'$ is the vector of predetermined variables, $\phi = \text{vec}(\Phi)$, $\Phi = (\mu, \Phi_{11}, \dots, \Phi_{r1}, \dots, \Phi_{r1}, \dots, \Phi_{rp})'$, $\otimes$ denotes the Kronecker product and vec the column-wise vectorization operator that stacks the column of a matrix in a column vector.

Then we assume a set of dependent infinite mixture priors G_i , $i = 1, \dots, r$, for the unknown parameters, ϕ and Σ , and obtain the following conditional distribution of Y_{it} (given X_t)

$$f_{it}(y) = \sum_{k \geq 1} W_{ik} \mathcal{K}_t(y|\tilde{\psi}_{ik}, X_t) \quad i = 1, \dots, r \quad (10)$$

where

$$\mathcal{K}_t(y|\psi, X_t) = (2\pi)^{-m/2} |\Sigma|^{-1/2} \exp \left\{ (y - (I_m \otimes X'_t) \phi)' \right. \\ \left. \times \Sigma^{-1} (y - (I_m \otimes X'_t) \phi) \right\} \quad (11)$$

is a multivariate Gaussian kernel of parameter $\psi = (\phi, \Sigma)$. In this infinite mixture representation the sources of conditional dependence between two units, Y_{it} and Y_{jt} , are the random weights, $(W_{ik})_k$ and $(W_{jk})_k$, and the random atoms $(\tilde{\psi}_{ik})_k$ and $(\tilde{\psi}_{jk})_k$.

3. Beta-product dependent Pitman–Yor processes

In this section we propose a new class of dependent stick-breaking processes in such a way that each marginal prior random measure G_i is a Pitman–Yor process. This result follows from the Sethuraman representation (1) if one considers a multivariate distribution for $(S_{1k}, \dots, S_{rk})$ such that $S_{ik} \sim \text{Beta}(1 - \alpha, \theta_i + \alpha k)$ for every i and k .

Note that there are many possible definitions of multivariate beta distribution (e.g. McKenzie (1985); Olkin and Liu (2003); Nadarajah and Kotz (2005); Taddy (2010) and Trippa et al. (2011)), but not all of them have a tractable stochastic representation and lead to simple Bayesian inference procedures. For this reason we follow Nadarajah and Kotz (2005) and consider a suitable product of independent beta random variables. More specifically we apply the following result.

Proposition 1 (Radhakrishna Rao (1949)). If $U_1, U_2, \dots, U_p$ are independent beta random variables with shape parameters (a_i, b_i) , $i = 1, 2, \dots, p$ and if $a_{i+1} = a_i + b_i$, $i = 1, 2, \dots, p-1$, then the product $U_1 U_2 \dots U_p$ is a beta random variable with parameters $(a_1, b_1 + \dots + b_p)$.

We propose two alternative specifications of the multidimensional beta variables. Specifically, if we set

$$(S_{1k}, S_{2k}, \dots, S_{rk}) = (V_{0k}V_{1k}, V_{0k}V_{2k}, \dots, V_{0k}V_{rk}) \quad (12)$$

with $V_{0k}, \dots, V_{rk}$ independent, $V_{ik} \sim \text{Beta}(\theta_{0k} + \theta_{1k}, \theta_{2k})$, $i = 1, 2, \dots, r$, and $V_{0k} \sim \text{Beta}(\theta_{0k}, \theta_{1k})$, then $S_{ik} \sim \text{Beta}(\theta_{0k}, \theta_{1k} + \theta_{2k})$.

As an alternative we consider

$$(S_{1k}, S_{2k}, \dots, S_{rk}) = (V_{0k}V_{1k} \dots V_{r-1k}, V_{0k}V_{1k} \dots V_{r-2k}, \dots, V_{0k}) \quad (13)$$

with $V_{0k}, \dots, V_{r-1k}$ independent and $V_{ik} \sim \text{Beta}(\theta_{0k} + \dots + \theta_{ik}, \theta_{i+1k})$, $i = 0, \dots, r-1$, that gives $S_{ik} \sim \text{Beta}(\theta_{0k}, \theta_{1k} + \dots + \theta_{r+1-i,k})$.

If (12) holds, from Lemma 1 in Ishwaran and James (2001), the process $(G_1, \dots, G_r)$ is well-defined, i.e. $\sum_{k \geq 1} W_{ik} = 1$ a.s. for every $i = 1, \dots, r$, when $\sum_{k \geq 1} \log(1 + \theta_{0k}/(\theta_{1k} + \theta_{2k})) = +\infty$. Analogously, assuming (13), the process is well-defined when $\sum_{k \geq 1} \log(1 + \theta_{0k}/\theta_{1k}) = +\infty$.

3.1. Pitman–Yor process marginal prior

In this section we show that by a suitable choice of the parameters in (12)–(13) we obtain a vector of dependent random measures with Pitman–Yor marginal prior distributions.

Consider (12) and set $\theta_{0k} = 1 - \alpha$, $\theta_{1k} = \theta_1$ and $\theta_{2k} = \theta_2 + \alpha k$, with $\theta_1 > 0$, $\theta_2 > 0$ and $0 \leq \alpha < 1$. Since the random variables $V_{0k}, \dots, V_{rk}$ are independent and

$$\begin{aligned} V_{0k} &\sim \text{Beta}(1 - \alpha, \theta_1), \\ V_{ik} &\sim \text{Beta}(1 - \alpha + \theta_1, \theta_2 + \alpha k) \quad i = 1, \dots, r, \end{aligned} \quad (14)$$

Proposition 1 yields that $S_{ik} \sim \text{Beta}(1 - \alpha, \theta_1 + \theta_2 + \alpha k)$.

Alternatively, in (13) set $\theta_{0k} = 1 - \alpha$, $\theta_{1k} = \theta_1 + \alpha k$, and $\theta_{ik} = \theta_i$ for $i \geq 2$ with $\theta_i > 0$ and $0 \leq \alpha < 1$. Since, in this case,

$$\begin{aligned} V_{0k} &\sim \text{Beta}(1 - \alpha, \theta_1 + \alpha k), \\ V_{1k} &\sim \text{Beta}(1 + \theta_1 + \alpha(k-1), \theta_2), \dots, \\ \dots V_{r-1k} &\sim \text{Beta}(1 + \theta_1 + \dots + \theta_{r-1} + \alpha(k-1), \theta_r), \end{aligned} \quad (15)$$

then $S_{ik} \sim \text{Beta}(1 - \alpha, \theta_1 + \dots + \theta_{r-i+1} + \alpha k)$.

Summarizing, if G_{0i} denotes the distribution of $\tilde{\psi}_{ik}$, we have proved the following

Proposition 2. If (12) and (14) hold true, then G_i is a PY($\alpha, \theta_1 + \theta_2, G_{0i}$) for every $i = 1, \dots, r$. If (13) and (15) hold true, then G_i is a PY($\alpha, \theta_1 + \dots + \theta_{r-i+1}, G_{0i}$) for every $i = 1, \dots, r$.

It is worth noticing that the idea of using product of beta random variables for the weights of a dependent Dirichlet process is not new. Similar constructions are detailed and applied in Taddy (2010) and Griffin and Steel (2011). Taddy (2010) is perhaps closest in spirit to our dependent PY process. The author applies the McKenzie (1985) autoregressive beta random process, to define the following autoregressive beta stick-breaking process: $S_{1k} \sim \text{Beta}(a, b)$ for all $k \geq 1$, and

$$S_{jk} = 1 - U_{jk}(1 - R_{jk}S_{j-1,k}),$$

for $j > 1$, where U_{jk} are i.i.d. $\text{Beta}(b, a - \rho)$ and R_{jk} are i.i.d. $\text{Beta}(\rho, a - \rho)$. In this way S_{jk} turns out to be $\text{Beta}(a, b)$ for all j and k , and for $a = 1$ one gets Dirichlet process marginal priors. The motivations, given in Taddy (2010), for this autoregressive

structure are that it allows for explicit prior correlation structure to model parameters and that it can be estimated sequentially with a simple particle filter algorithm. Although not discussed in Taddy (2010), one can easily modify his construction in order to obtain Pitman–Yor marginal priors by setting $a = 1 - \alpha$ and $b = b_k = \theta + \alpha k$. In this case S_{jk} turns out to be $\text{Beta}(1 - \alpha, \theta + \alpha k)$ variables for all $j \geq 1$.

In this paper, we choose the different specifications (14) and (15) mainly for two reasons. First, in (14) all possible pairs (S_{jk}, S_{ik}) have the same correlation, while the correlation in the Taddy (2010)'s construction is decreasing in $|j - i|$. While the decreasing correlation property seems reasonable for modelling the cluster dynamics over time, its application to panel or repeated measurement models may be restrictive. In these models, the correlation structure between units or group of observations is not always related to $|j - i|$ and in the absence of prior information on the dependence structure it seems preferable to use (14), which allows for both identical marginal prior distributions and the correlation level between the clustering features of the different units. Secondly, with (15) one obtains Pitman–Yor marginal prior distributions with different parameters, while extending (Taddy, 2010) autoregressive construction one obtains the same Pitman–Yor marginals for all the components. Finally, the stochastic representations (14) and (15) allow the derivation of a slice sampling algorithm in the spirit of Walker (2007) and Kalli et al. (2011). As a side remark note that for constructing an r -dimensional vector of weights (Taddy, 2010) uses $2r - 1$ latent variables while (14) and (15) require $r + 1$ and r respectively.

For the sake of simplicity in what follows we will mainly consider $r = 2$. According to the above construction schemes, the two alternative specifications of (S_{1k}, S_{2k}) are:

- (H1) $(S_{1k}, S_{2k}) = (V_{0k}V_{1k}, V_{0k}V_{2k})$, with V_{0k}, V_{1k}, V_{2k} independent, $V_{0k} \sim \text{Beta}(1 - \alpha, \theta_1)$ and $V_{ik} \sim \text{Beta}(1 - \alpha + \theta_1, \theta_2 + \alpha k)$, $i = 1, 2$, where $\theta_1 > 0$, $\theta_2 > 0$ and $\alpha \in [0, 1]$;
- (H2) $(S_{1k}, S_{2k}) = (V_{0k}V_{1k}, V_{0k})$, with V_{0k}, V_{1k} independent, $V_{0k} \sim \text{Beta}(1 - \alpha, \theta_1 + \alpha k)$ and $V_{1k} \sim \text{Beta}(1 + \theta_1 + \alpha(k-1), \theta_2)$ with $\theta_1 > 0$, $\theta_2 > 0$ and $\alpha \in [0, 1]$.

Since with this construction the G_i 's are Pitman–Yor processes, we call (G_1, G_2) Beta-Product Dependent Pitman–Yor Process, $\beta_i - \text{DPY}(\varphi, G_0)$ for short, of parameters $\varphi = (\theta_1, \theta_2, \alpha)$ and base measure G_0 , where $i = 1$ for (H1) and $i = 2$ for (H2). If one takes $\alpha = 0$, the resulting process has Dirichlet marginal prior distributions, in this case we shall write $\beta_i - \text{DD}(\varphi, G_0)$.

It should be noted that the two processes have different marginal behaviours. The $\beta_1 - \text{DPY}(\varphi, G_0)$ process has marginal prior distributions with the same precision parameter and thus should be used as a prior when the clustering is expected to be similar along the different vector components. In the $\beta_2 - \text{DPY}(\varphi, G_0)$ process, the precision parameter decreases along the vector dimension. This process should be used when a priori one expects that the clustering features are different across units.

We conclude this section with a few remarks on the limiting cases that can help in the interpretation of the parameters. Under (H1), (S_{1k}, S_{2k}) converges (in distribution) to (V_{1k}, V_{2k}) as $\theta_1 \rightarrow 0$, where V_{1k} and V_{2k} are independent random variables with distribution $\text{Beta}(1 - \alpha, \theta_2 + \alpha k)$. While, under (H2), (S_{11}, S_{21}) converges to (V_0, V_0) as $\theta_2 \rightarrow 0$, where V_0 is a $\text{Beta}(1 - \alpha, \theta_1 + \alpha k)$ random variable. In particular, if one assumes (H2) and $\tilde{\psi}_{1k} = \tilde{\psi}_{2k}$ for all k , when $\theta_2 \rightarrow 0$, one gets the limiting case in which all the observations are sampled from a common mixture of Pitman–Yor processes. In other words, in this limiting case, one considers the observations (globally) exchangeable, so no distinction between the two blocks are allowed. The other limiting case is when one assumes (H1) and takes $(\tilde{\psi}_{1k}, \tilde{\psi}_{2k})$ to be independent random elements with probability distributions G_{01} and G_{02} . In the limit

for $\theta_1 \rightarrow 0$, one obtains two independent Pitman–Yor processes G_1 and G_2 with base measures G_{01} and G_{02} . With this choice, the observations are grouped into two independent blocks which do not share information.

3.2. Dirichlet process marginal prior

For $\alpha = 0$ in (14) and (15) one gets Dirichlet process marginal prior distributions and, for this special case, it is possible to find an explicit formula for the correlation between G_i and G_j as a function of the parameters $(\theta_1, \theta_2, \dots)$. These results can be used for parameter elicitation purposes.

Let us assume time- and unit-independent kernels $\mathcal{K}(y|\psi)$, and define, for every $y \in \mathbb{Y}$ and every $i, j = 1, \dots, r$ with $i \neq j$,

$$\begin{aligned}\kappa_{G_{0i}}(y) &= \int \mathcal{K}(y|\psi) G_{0i}(d\psi), & \kappa_{G_{0j}}(y) &= \int \mathcal{K}(y|\psi) G_{0j}(d\psi), \\ \kappa_{G_{0ij}}(y) &= \int \mathcal{K}(y|\psi_i) \mathcal{K}(y|\psi_j) G_{0ij}(d\psi_i d\psi_j)\end{aligned}$$

where G_{0i} denotes the distribution of $\tilde{\psi}_{i1}$ and G_{0ij} the distribution of $(\tilde{\psi}_{i1}, \tilde{\psi}_{j1})$.

Proposition 3. Let (12) and (14) or (13) and (15) hold with $\alpha = 0$, then for all measurable sets A and B and every y in $\mathbb{Y}$

$$\begin{aligned}\text{Cor}(G_i(A), G_j(B)) \\ = C_{ij} \times \frac{G_{0ij}(A \times B) - G_{0i}(A)G_{0j}(B)}{\sqrt{G_{0i}(A)(1 - G_{0i}(A))G_{0j}(B)(1 - G_{0j}(B))}}\end{aligned}\quad (16)$$

$$\begin{aligned}\text{Cor}(f_i(y), f_j(y)) \\ = C_{ij} \times \frac{\kappa_{G_{0ij}}(y) - \kappa_{G_{0i}}(y)\kappa_{G_{0j}}(y)}{\sqrt{\kappa_{G_{0i}}(y)(1 - \kappa_{G_{0i}}(y))\kappa_{G_{0j}}(y)(1 - \kappa_{G_{0j}}(y))}}\end{aligned}\quad (17)$$

with

$$C_{ij} = \frac{(1 + \theta_1 + \theta_2)(1 + \theta_1)}{(1 + \theta_1)(1 + \theta_1 + \theta_2) + \theta_2}$$

under (12) and (14) and

$$C_{ij} = \frac{2\sqrt{(1 + \theta_1 + \dots + \theta_{r-i+1})(1 + \theta_1 + \dots + \theta_{r-j+1})^{\frac{3}{2}}}}{2(1 + \theta_1 + \dots + \theta_{r-j+1})^2 + (2 + \theta_1 + \dots + \theta_{r-j+1})(\theta_{r-j+2} + \dots + \theta_{r-i+1})}$$

under (13) and (15).

Note that the correlation between random measures and densities, given in the above proposition, has two components. The first one is C_{ij} which is affected by the correlation level between the stick variables (see proof of Proposition 3 in Appendix A), and the second one is given by the remaining term which reflects the dependence level between the atoms. In particular, if one assumes that the atoms are common to all G_i (see Eq. (18) in the next section) then $\text{Cor}(G_i(A), G_j(A)) = C_{ij}$, which suggests a possible interpretation for C_{ij} as a correlation and highlights the contribution of the stick variables to the correlation between random measures. We will discuss the assumptions on the atoms in the next section and focus, in the rest of this section, on the effects of the stick-variables correlation on C_{ij} , for different choices of the parameters. Assuming for simplicity $r = 2$, in the cases $\alpha = 0$ one obtains the following correlation between S_{1h} and S_{2h}

$$\text{Cor}(S_{11}, S_{21}) = \begin{cases} \frac{\theta_1(2 + \theta_1 + \theta_2)}{(2 + \theta_1)(\theta_1 + \theta_2)} & \text{for (H1)} \\ \sqrt{\frac{\theta_1(2 + \theta_1 + \theta_2)}{(2 + \theta_1)(\theta_1 + \theta_2)}} & \text{for (H2).} \end{cases}$$

Fig. 1 shows the correlation level between the stick-breaking components and the random measures for different values

of θ_1 and θ_2 . According to the left graphs of Fig. 1, the two parameterizations allow for covering the whole range of possible correlation values in the unit interval. For instance a low correlation between the components of the stick-breaking corresponds to low values of θ_1 , say between 0 and 0.1, for any choice of θ_2 . The right graphs of Fig. 1 show the effect of θ_1 and θ_2 on C_{12} .

4. A hierarchical structure for the atoms

The simplest assumption for the atoms is that they are common to all the measures G_i . Otherwise stated this means that

$$(\tilde{\psi}_{1k}, \dots, \tilde{\psi}_{rk}) = (\tilde{\psi}_{0k}, \tilde{\psi}_{0k}, \dots, \tilde{\psi}_{0k}) \quad (18)$$

with $\tilde{\psi}_{0k}$ distributed according to a given common probability measure on Ψ . Although in some situations it is reasonable to assume that the components of the mixture are the same for all the units, the shared atom simplification (18) is a quite severe restriction when the units (or sub-samples) exhibit a high degree of heterogeneity. For this reason, we consider a more elaborate hierarchical structure, that could include covariates (or exogenous effects) related to the specific block i in the law of $\tilde{\psi}_{ik}$.

4.1. ANOVA-like repeated measure models

Let us consider the Gaussian repeated measure model introduced in Eqs. (6)–(7), Section 2.2.1. The components of the parameter $\psi = (\mu, \sigma^2)$ represent a mean and a variance respectively. To allow for various degrees of pooling of information across units, one can assume for the parameters of the i th kernel the ANOVA-like scheme of De Iorio et al. (2004), i.e. $\psi = (\mu_0 + \mu_i, \sigma_0^2 \sigma_i^2)$. With this choice, each atom is the vector

$$\tilde{\psi}_{ik} = (\tilde{\mu}_{0k} + \tilde{\mu}_{ik}, \tilde{\sigma}_{0k}^2 \tilde{\sigma}_{ik}^2) \quad (19)$$

where $\tilde{\mu}_{0k}, \tilde{\mu}_{1k}, \dots, \tilde{\mu}_{rk}$ and $\tilde{\sigma}_{0k}^2, \tilde{\sigma}_{1k}^2, \dots, \tilde{\sigma}_{rk}^2$ are independent random variables. In this case $\tilde{\mu}_{0k}$ and $\tilde{\sigma}_{0k}^2$ represent the common mean and variance (of the k -th mixture component) and $\tilde{\mu}_{ik}$ and $\tilde{\sigma}_{ik}^2$ the i th factor-specific mean and variance (of the k -th mixture component).

We complete the hierarchical specification of the base measure G_0 by setting

$$(\tilde{\mu}_{ik}, \tilde{\sigma}_{ik}^2) \sim \mathcal{N}(0, s_i^{-2}) \mathcal{IG}(\lambda/2, \lambda/2) \quad i = 1, 2$$

$$(\tilde{\mu}_{0k}, \tilde{\sigma}_{0k}^2) \sim \mathcal{N}(0, s_0^{-2}) \mathcal{IG}(\varepsilon/2, \varepsilon/2)$$

where $\mathcal{N}(0, s^{-2})$ denotes a Gaussian distribution of mean 0 and precision s^2 , and $\mathcal{IG}(a, b)$ denotes an inverse gamma distribution of parameters (a, b) .

4.2. Hierarchical Bayesian VAR models

In the panel VAR model given in Eq. (8) the number of parameters, $(N^2p + N + N(N + 1)/2)$ with $N = mr$, increases rapidly with the number of units and variables possibly leading to an overfitting problem. Our Bayesian non-parametric approach deals with the overfitting problem through a suitable choice of the base measure of the Pitman–Yor process prior. We follow here a hierarchical specification of the base measure which can be used to incorporate cross-equation interdependences and various degrees of information pooling across units (e.g., see Chib and Greenberg (1995) and Min and Zellner (1993)).

For the i th unit, the first stage equation of our hierarchical model is given by Eq. (9). As regards the second stage of hierarchy, our prior setting is

$$\tilde{\psi}_{ik} = (\tilde{\phi}_{0k} + \tilde{\phi}_{ik}, \tilde{\sigma}_{0k} \tilde{\Sigma}_{ik})$$

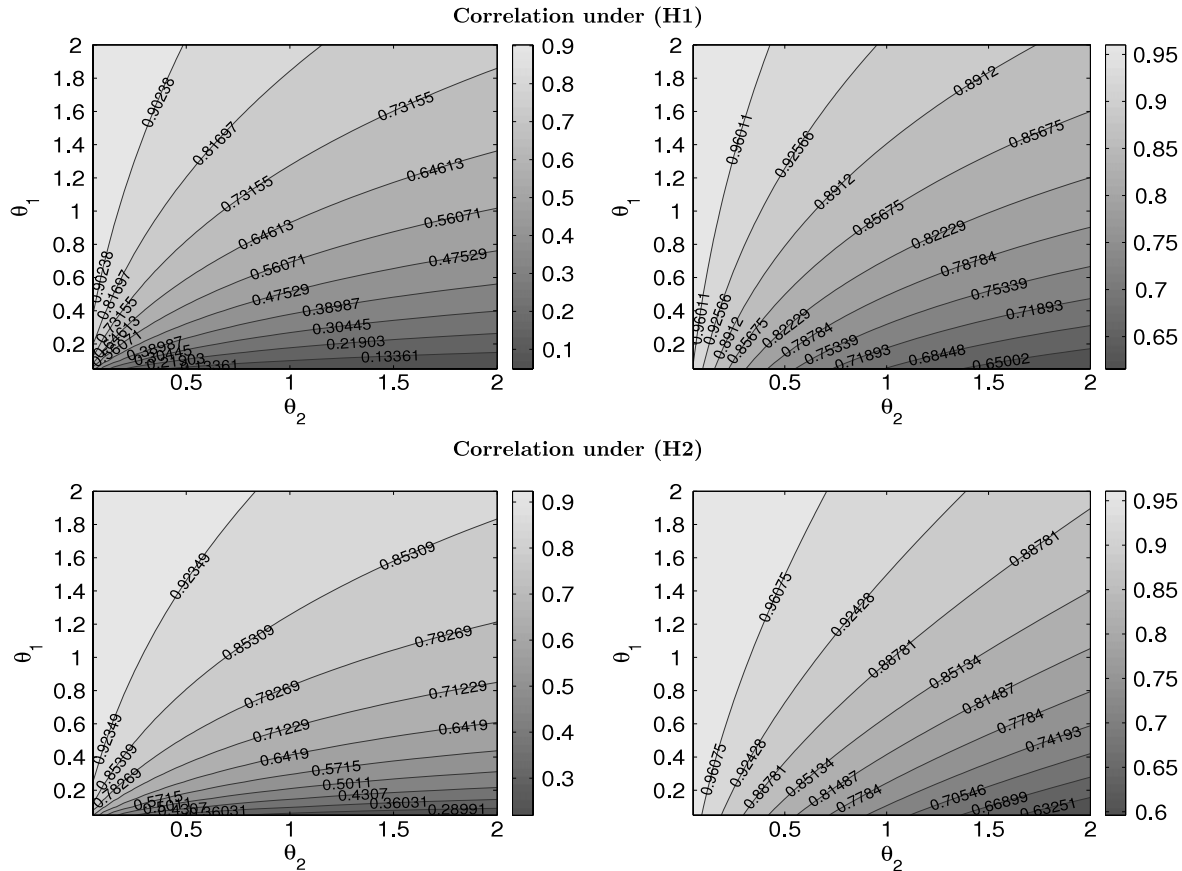


Fig. 1. Left column: correlation between S_{11} and S_{21} under (H1) (first row) and (H2) (second row). Right column: C_{12} , that is correlation between G_1 and G_2 , assuming common atoms, under (H1) (first row) and (H2) (second row). In each graph, white is used for correlation values equal to one, black for correlation values equal to zero and grey levels for the other correlation values in the unit interval.

$i = 1, \dots, r$, where $\tilde{\phi}_{0k}$ and $\tilde{\sigma}_{0k}$ are common factors to all countries, while $\tilde{\phi}_{ik}$ and $\tilde{\Sigma}_{ik}$ are country-specific factors. Above $\tilde{\phi}_{0k}$, $\tilde{\sigma}_{0k}$, $\tilde{\phi}_{ik}$ and $\tilde{\Sigma}_{ik}$ are independent and

$$\begin{aligned} (\tilde{\phi}_{ik}, \tilde{\Sigma}_{ik}^{-1}) &\sim \mathcal{N}_m(0, \tilde{\gamma}_{ik}) \mathcal{W}_2(\lambda, \Lambda) \quad i = 1, 2 \\ (\tilde{\phi}_{0k}, \tilde{\sigma}_{0k}^{-1}) &\sim \mathcal{N}_m(0, \gamma_0) \mathcal{G}(\varepsilon/2, \varepsilon/2) \end{aligned} \quad (20)$$

where $m = 2(2rp + 1)$, $\mathcal{W}_2(\lambda, \Lambda)$ denotes a bivariate Wishart distribution of parameters λ and Λ and $\mathcal{G}(\varepsilon/2, \varepsilon/2)$ denotes the gamma distribution of parameters $\varepsilon/2$ and $\varepsilon/2$. A third stage of the hierarchy is imposed by specifying a prior also for all the $\tilde{\gamma}_{ik}$, $i = 1, 2, k \geq 1$. A specific choice of such a prior will be presented in Section 6.2.

5. Slice sampling algorithm for posterior simulation

For posterior computation, we propose an extension of the slice sampling algorithm introduced in Walker (2007) and Kalli et al. (2011). For the sake of simplicity we shall describe the sampling strategy for a vector of Beta-Product DPY with $r = 2$. The proposed algorithm can be easily extended to the case $r > 2$.

Recall that the stick variables in the β_i -DPY(φ, G_0) are defined by

$$(S_{1k}, S_{2k}) = (V_{0k}V_{1k}, V_{0k}V_{2k})$$

for a sequence of independent vectors $V_k = (V_{0k}, V_{1k}, V_{2k})$, with the convention $V_{2k} = 1$ and $V_k = (V_{0,k}, V_{1,k})$ under (H2).

In order to deal in the same time with both the repeated measure model of Section 2.2.1 and the panel VAR model of Section 2.2.2, we assume that for any $t \geq 1$ the conditional

distribution of Y_{it} given $Z_t = [Y_{is} : i = 1, 2, s = 1, \dots, t-1]$ is

$$f_{it}(y) = f_{it}(y|Z_t) = \sum_{k \geq 1} W_{ik} \mathcal{K}_t(y|\tilde{\psi}_{ik}, Z_t) \quad i = 1, 2 \quad (21)$$

where $\mathcal{K}_t$ is a suitable kernel depending on some parameter $\tilde{\psi}$ and possibly on t and Z_t . As we have seen, both infinite mixtures of repeated measurement models (6)–(7) and infinite mixtures of panel VAR models (10)–(11) can be written in this way.

Starting from (21), the key idea of the slice sampling is to find a finite number of variables to be sampled. To this end one introduces the allocation variables D_{it} 's for the observations Y_{it} and the slice variables U_{it} 's (taking values on $[0, 1]$) for the weights $W_{i,D_{it}}$. The joint density of (Y_{it}, U_{it}, D_{it}) given $[Z_t, (U_{js}, D_{js}) : j = 1, 2, s = 1, \dots, t-1]$ is

$$f_{it}(y, u, d) = \mathbb{I}\{u \leq W_{i,d}\} \mathcal{K}_t(y|\tilde{\psi}_{id}, Z_t). \quad (22)$$

With this position, the marginal density of Y_{it} given Z_t is (21) and the likelihood function for the augmented variables is available as a simple product of terms.

We shall use the notation

$$\begin{aligned} Y_i^{(T_1)} &= (Y_{i1}, \dots, Y_{iT_1}), & D_i^{(T_1)} &= (D_{i1}, \dots, D_{iT_1}), \\ U_i^{(T_1)} &= (U_{i1}, \dots, U_{iT_1}) \end{aligned}$$

and write: $\tilde{\psi}$ for $(\tilde{\psi}_k)_k$, V for $(V_k)_k$, U for $[U_1^{(T_1)}, U_2^{(T_2)}]$, D for $[D_1^{(T_1)}, D_2^{(T_2)}]$ and Y for $[Y_1^{(T_1)}, Y_2^{(T_2)}]$.

In the applications we further assume a prior for $\varphi = (\theta_1, \theta_2, \alpha)$. Following the standard practice for these Bayesian non-parametric models, we let $\tilde{\psi}$ be independent of $\tilde{\varphi} = (\tilde{\theta}_1, \tilde{\theta}_2, \tilde{\alpha})$,

while the distribution of V_j depends on $\tilde{\varphi}$ through (H1) or (H2), so we shall write $P\{V_j \in dv_j | \tilde{\varphi}\}$.

In order to describe the structure of the Gibbs sampler we introduce some more notation. Define, for $i = 1, 2$ and $k \geq 1$, the set of indexes of the observations allocated to the k -th component of the mixture, $\mathcal{D}_{i,k} = \{t \in \{1, \dots, T_i\} : D_{i,t} = k\}$, and the set of indexes of the non-empty mixture components, $\mathcal{D}^* = \{k : \mathcal{D}_{1,k} \cup \mathcal{D}_{2,k} \neq \emptyset\}$. In our MCMC algorithm we shall treat $V = (V^*, V^{**})$ as two blocks of random length:

$$V^* = \{V_k : k \in \mathcal{D}^*\}, \quad V^{**} = \{V_k : k \notin \mathcal{D}^*\}.$$

The Gibbs sampler simulates iteratively the following blocks of parameters and latent variables:

- (i) $\tilde{\psi}$ given $[V, U, D, \tilde{\varphi}, Y]$;
- (ii) $[V, \tilde{\varphi}]$ given $[\tilde{\psi}, U, D, Y]$ by exploiting the decomposition:
 - (ii.a) $[V^*, \tilde{\varphi}]$ given $[D, \tilde{\psi}, Y]$,
 - (ii.b) V^{**} given $[D, V^*, \tilde{\psi}, \tilde{\varphi}, Y]$;
- (iii) D given $[V, U, \tilde{\psi}, \tilde{\varphi}, Y]$.

All the full conditionals can be deduced from the joint distribution given in Eq. (33) in Appendix A. Further details on the MCMC algorithm proposed here are given in Appendix B.

6. Illustrations

6.1. DPY(φ, G_0) mixtures of Gaussian distributions

6.1.1. Estimation results

We consider DPY(φ, G_0) Gaussian mixture model for inference on synthetic data generated from finite Gaussian mixtures. For the atoms we assume the hierarchical structure described in Section 4.1.

We assume independent gamma prior distributions $\mathcal{G}(\zeta_{11}, \zeta_{21})$, $\mathcal{G}(\zeta_{12}, \zeta_{22})$ for the vector $\tilde{\theta} = (\tilde{\theta}_1, \tilde{\theta}_2)$ of precision parameters and uniform prior $\mathcal{U}_{[0,1]}$ for the parameter $\tilde{\alpha}$. In summary the Bayesian non-parametric model is

$$\begin{aligned} Y_{it} | (\mu_{it}^*, \sigma_{it}^{*2}) &\stackrel{\text{ind}}{\sim} \mathcal{N}(\mu_{it}^*, \sigma_{it}^{*2}) \quad i = 1, 2, t \geq 1 \\ (\mu_{it}^*, \sigma_{it}^{*2}) | G_1, G_2 &\stackrel{\text{iid}}{\sim} G_i \quad i = 1, 2 \\ (G_1, G_2) | \tilde{\varphi} &\sim \beta - \text{DPY}(\tilde{\varphi}, G_0) \\ \tilde{\varphi} &\sim \mathcal{G}(\zeta_{11}, \zeta_{21}) \mathcal{G}(\zeta_{12}, \zeta_{22}) \mathcal{U}_{[0,1]}. \end{aligned}$$

Recall that, as described in Section 4.1, the base measure G_0 is such that

$$(\mu_{it}^*, \sigma_{it}^{*2}) = (\tilde{\mu}_{0D_{it}} + \tilde{\mu}_{iD_{it}}, \tilde{\sigma}_{0D_{it}}^2 + \tilde{\sigma}_{iD_{it}}^2),$$

where D_{it} is the allocation variable of the observation Y_{it} , and

$$\begin{aligned} (\tilde{\mu}_{ik}, \tilde{\sigma}_{ik}^2) &\sim \mathcal{N}(0, s_i^{-2}) \mathcal{I} \mathcal{G}(\lambda/2, \lambda/2) \quad i = 1, 2 \\ (\tilde{\mu}_{0k}, \tilde{\sigma}_{0k}^2) &\sim \mathcal{N}(0, s_0^{-2}) \mathcal{I} \mathcal{G}(\varepsilon/2, \varepsilon/2). \end{aligned}$$

Appendix B describes the sampling strategy in detail.

We simulate $T = 100$ independent vectors, $(Y_{1,t}, Y_{2,t})$ with $t = 1, \dots, T$, of observations. The components of the vectors $(Y_{1,t}, Y_{2,t})$ are independent and alternatively follow one of the following models.

- The same three-component mixture of normals (model Mix1)

$$\begin{aligned} Y_{1,t} &\sim \frac{1}{3} \mathcal{N}(-10, 1) + \frac{1}{3} \mathcal{N}(0, 1) + \frac{1}{3} \mathcal{N}(10, 1) \\ Y_{2,t} &\sim \frac{1}{3} \mathcal{N}(-10, 1) + \frac{1}{3} \mathcal{N}(0, 1) + \frac{1}{3} \mathcal{N}(10, 1). \end{aligned}$$

- The same three-component mixture of normals with different component probabilities (model Mix2)

$$\begin{aligned} Y_{1,t} &\sim \frac{1}{3} \mathcal{N}(-10, 1) + \frac{1}{3} \mathcal{N}(0, 1) + \frac{1}{3} \mathcal{N}(10, 1) \\ Y_{2,t} &\sim \frac{1}{6} \mathcal{N}(-10, 1) + \frac{4}{6} \mathcal{N}(0, 1) + \frac{1}{6} \mathcal{N}(10, 1). \end{aligned}$$

- Two different mixtures with two common components (model Mix3)

$$\begin{aligned} Y_{1,t} &\sim \frac{1}{4} \mathcal{N}(0, 0.5) + \frac{1}{4} \mathcal{N}(3, 0.25) + \frac{1}{4} \mathcal{N}(2, 0.25) \\ &\quad + \frac{1}{4} \mathcal{N}(5, 0.5) \\ Y_{2,t} &\sim \frac{1}{4} \mathcal{N}(0, 0.5) + \frac{1}{4} \mathcal{N}(3, 0.25) + \frac{1}{4} \mathcal{N}(-3, 0.25) \\ &\quad + \frac{1}{4} \mathcal{N}(7, 0.5). \end{aligned}$$

- Two different mixtures with two common modes (model Mix4) and small modes in one of the mixtures

$$\begin{aligned} Y_{1,t} &\sim \frac{2}{9} \mathcal{N}(-10, 0.4) + \frac{1}{12} \mathcal{N}(-5, 0.4) \\ &\quad + \frac{1}{12} \mathcal{N}(-3, 0.4) + \frac{2}{9} \mathcal{N}(0, 0.4) \\ &\quad + \frac{1}{12} \mathcal{N}(3, 0.4) + \frac{1}{12} \mathcal{N}(5, 0.4) + \frac{2}{9} \mathcal{N}(10, 0.4) \\ Y_{2,t} &\sim \frac{1}{3} \mathcal{N}(-10, 1) + \frac{1}{3} \mathcal{N}(0, 1) + \frac{1}{3} \mathcal{N}(10, 1). \end{aligned}$$

We apply the $\beta_i - \text{DD}(\varphi, H_0)$ and the $\beta_i - \text{DPY}(\varphi, H_0)$, $i = 1, 2$, on the different set of data. In the inference exercise, we choose a fairly non-informative prior specification for the mean and precision parameters of the base measure and set $s_i^2 = 0.1$, $\lambda = 0.5$ (see for example Walker (2007)). Similarly we assume $s_0^2 = 0.01$ and $\varepsilon = 1$. For the concentration parameters (θ_1, θ_2) of the stick-breaking components, we follow Kalli et al. (2011) and consider a prior with hyperparameters $(\zeta_{1j} = 0.01, \zeta_{2j} = 0.01)$, for $j = 1, 2$, in all the experiments from Mix1 to Mix4. This setting corresponds to diffuse prior distributions on the concentration parameters, with prior means $\mathbb{E}(\tilde{\theta}_1) = \mathbb{E}(\tilde{\theta}_2) = 1$ and variances $\mathbb{V}(\tilde{\theta}_1) = \mathbb{V}(\tilde{\theta}_2) = 100$. For each dataset and model, the Gibbs sampler, presented in the previous section, was run for 20,000 iterations. A burn-in period of 10,000 samples was discarded and at each Gibbs iteration from 10,000 onwards, a sample $(Y_{1,T+1}, Y_{2,T+1})$ from the predictive density was taken.

We found that dependent Dirichlet and Pitman–Yor process priors give different results, in terms of predictive densities and the posterior number of clusters, for datasets Mix3 and Mix4, while for Mix1 and Mix2 the two prior distributions give similar results in terms of predictive densities and have small differences on the number of clusters. For the expository purposes we show the graphical output only for the most interesting results, which we obtained for datasets Mix3 and Mix4. Right and left panels in Fig. 2 show the histograms of the two components (right and left columns of each panel) for each set of synthetic data (different rows). In the same figures the solid lines represent the estimated posterior predictive densities for $\beta_1 - \text{DD}(\varphi, H_0)$, $\beta_2 - \text{DD}(\varphi, H_0)$, $\beta_1 - \text{DPY}(\varphi, H_0)$ and $\beta_2 - \text{DPY}(\varphi, H_0)$. The approximated posterior of the number of clusters is given in Fig. 3.

The results for Mix3, in Fig. 2, show that $\beta_i - \text{DPY}(\varphi, H_0)$, $i = 1, 2$ have similar posterior predictive densities and $\beta_1 - \text{DPY}(\varphi, H_0)$ is doing better than $\beta_1 - \text{DD}(\varphi, H_0)$ in terms of the number of clusters (see also Fig. 3). DD and DPY under (H2) give similar results. For dataset Mix4 DD and DPY give different results both in terms of predictive densities and the posterior number of

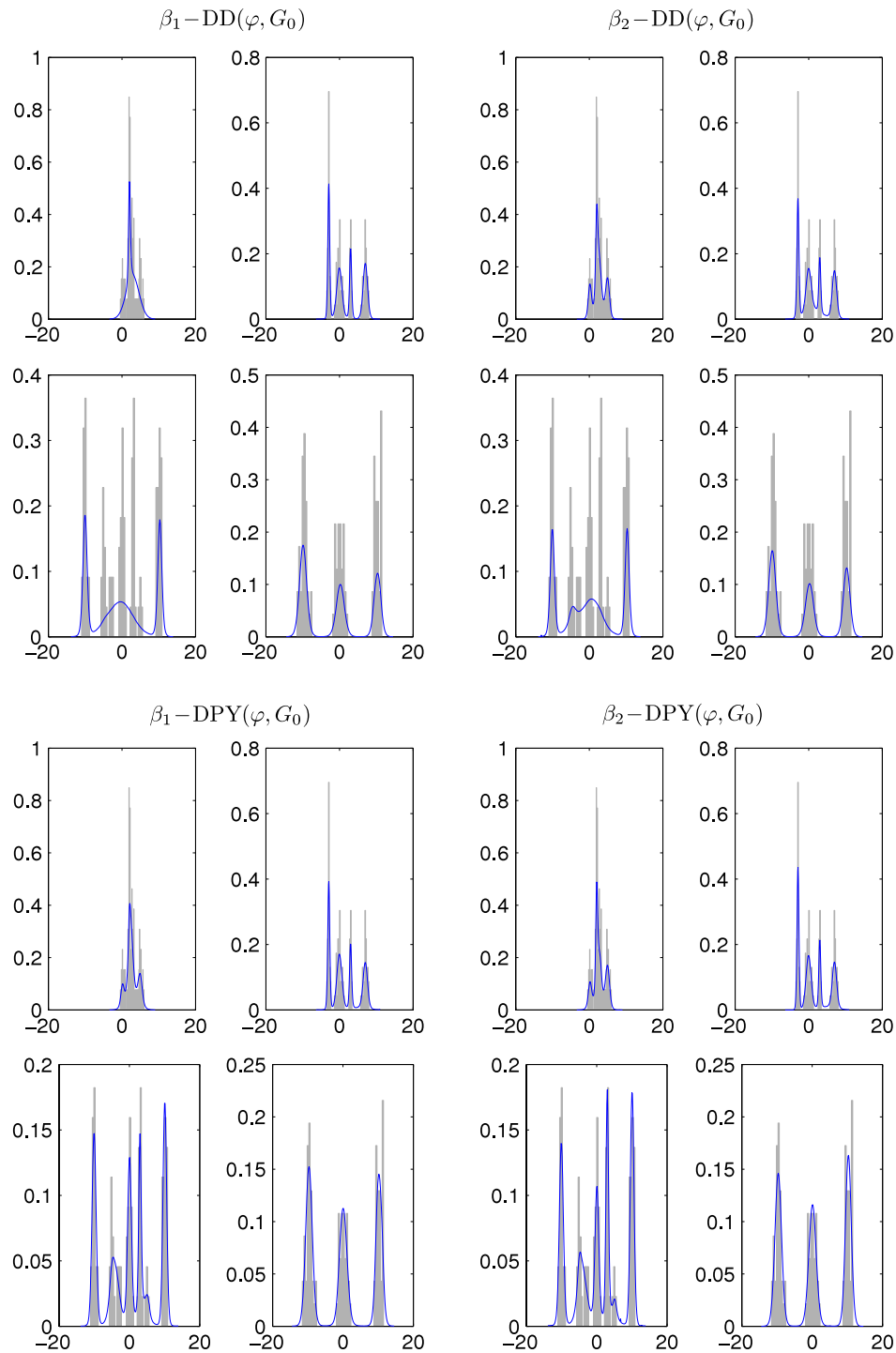


Fig. 2. Predictive distributions for different prior settings (panels $\beta_1 - \text{DD}(\varphi, G_0)$, $\beta_2 - \text{DD}(\varphi, G_0)$, $\beta_1 - \text{DPY}(\varphi, G_0)$ and $\beta_2 - \text{DPY}(\varphi, G_0)$). In each panel, data histograms for the first (left column) and second (right column) component and for the Mix3 (first row) and Mix4 (second row) dataset.

clusters. $\beta_i - \text{DD}(\varphi, H_0)$, $i = 1, 2$, are largely under-estimating the number of clusters while $\beta_i - \text{DPY}(\varphi, H_0)$, $i = 1, 2$, are doing better, especially $\beta_2 - \text{DPY}(\varphi, H_0)$.

6.1.2. Prior selection and sensitivity analysis

The results described in Section 6.1.1 are confirmed in the statistical comparison between the different non-parametric models. In the comparison we consider a Bayesian model averaging approach. We focus on the predictive densities of the proposed

models, due to the relevance of the density forecasts in many economics applications. Although there are many alternative ways to assign to the models a numerical score based on the ability of the predictive density and to predict the variables of interest (e.g., see [Gneiting and Raftery \(2007\)](#)), we follow [Jensen and Maheu \(2010\)](#) and apply the model pooling approach of [Geweke and Amisano \(2010\)](#) based on the log predictive score. Our set of predictive densities is $f(Y_t|Y_1, \dots, Y_{t-1}, M_j)$, $j = 1, \dots, J$, where $Y'_t = (Y'_{1t}, Y'_{2t})$ and M_j indicates model $\beta_1 - \text{DD}(\varphi, G_0)$ ($j = 1$),

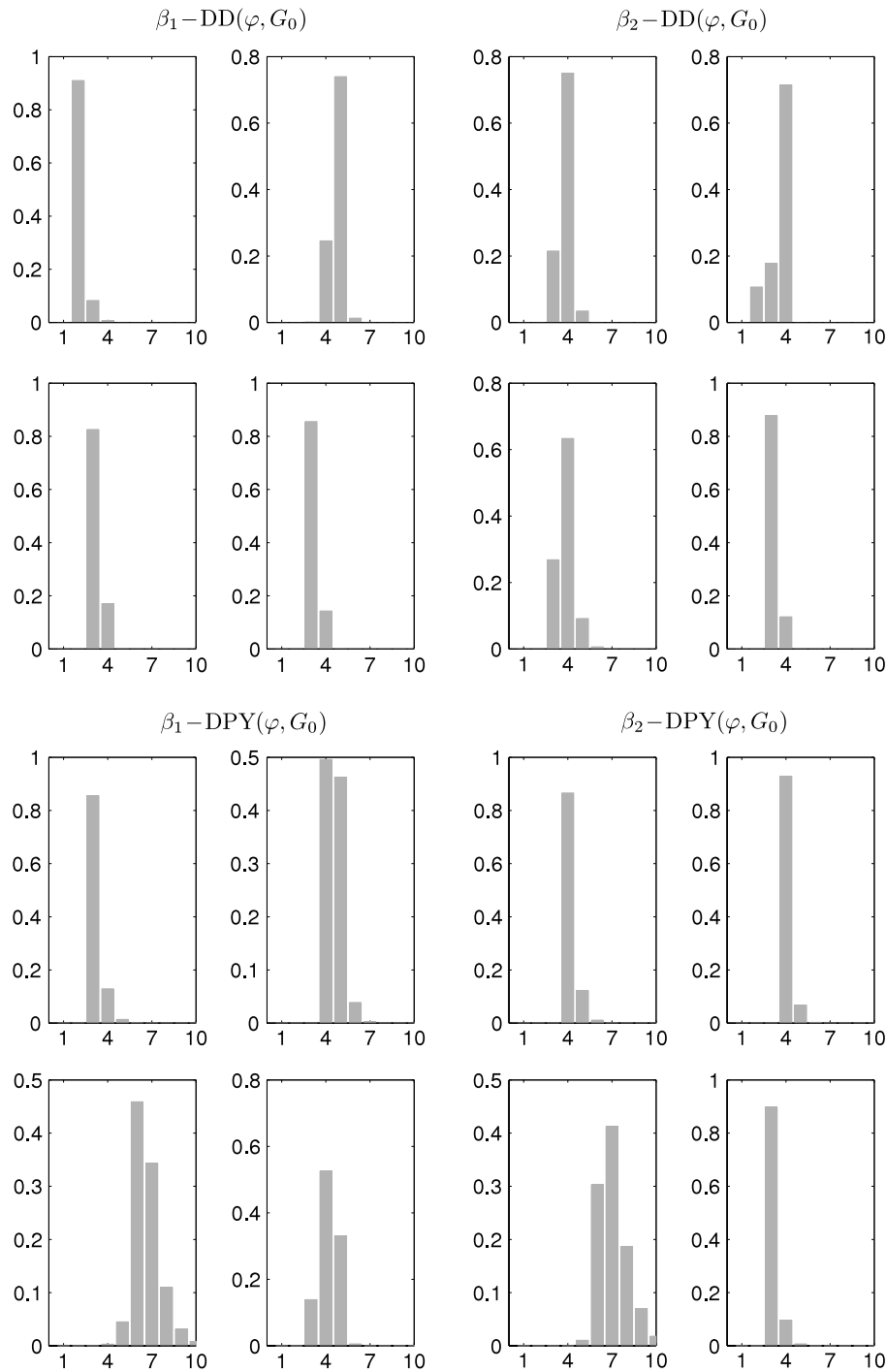


Fig. 3. Number of clusters for different prior settings (panels $\beta_i - DD(\varphi, G_0)$, $i = 1, 2$ and $\beta_i - DPY(\varphi, G_0)$). In each panel, histograms for the first (left column) and second component (right column) and for the Mix3 (first row) and Mix4 (second row) dataset.

$\beta_2 - DD(\varphi, G_0)$ ($j = 2$), $\beta_1 - DPY(\varphi, G_0)$ ($j = 3$) and $\beta_2 - DPY(\varphi, G_0)$ ($j = 4$). We define the combined predictive density

$$f(Y_t | Y_1, \dots, Y_{t-1}) = \sum_{j=1}^J w_j f(Y_t | Y_1, \dots, Y_{t-1}, M_j)$$

with $w_j \geq 0$ and $\sum_{j=1}^J w_j = 1$ and choose the combination weights $w = (w_1, \dots, w_J)$ to maximize the log pooled predictive score function

$$\max_w \sum_{t=\tau_1}^{\tau_2} \log \sum_{j=1}^J w_j f(Y_t | Y_1, \dots, Y_{t-1}, M_j).$$

The model-specific predictive densities are not available in analytical form, but can be approximated by using the Gibbs sampling scheme given in the previous section. In this application, for a given dataset, we set $\tau_1 = 50$ and $\tau_2 = 100$ and for each model M_j , $j = 1, \dots, 4$, we use 5000 MCMC draws to approximate the predictive density $f(Y_t | Y_1, \dots, Y_{t-1}, M_j)$ at the data point Y_t . We repeated the procedure for $t = \tau_1, \dots, \tau_2$ and obtain a sequence of densities for each model that can be used to find the optimal combination weights.

The results in Table 1 show that for all datasets DPY models have the better predictive ability. More specifically for Mix1 $\beta_1 - DPY(\varphi, H_0)$ has the highest combination weight, while for Mix2

Table 1

In each row: the simulation experiment (first column), the optimal pooled log-predictive score function (second column) and the associated optimal combination weights (remaining columns) for $\beta_1 - DD(w_1)$, $\beta_2 - DD(w_2)$, $\beta_1 - DPY(w_3)$ and $\beta_2 - DPY(w_4)$.

Dataset	Log-score	w_1	w_2	w_3	w_4
Mix1	−250.25	0.0000	0.0000	0.9087	0.0913
Mix2	−179.61	0.2256	0.0000	0.7744	0.0000
Mix3	−265.00	0.4674	0.0000	0.5326	0.0000
Mix4	−273.25	0.0000	0.0790	0.3827	0.5383

and Mix3 both DD and DPY under assumption (H1) are performing better than DD and DPY under assumption (H2). Although, for both datasets $\beta_1 - DPY(\varphi, H_0)$ has the highest combination weight. Finally on dataset Mix 4, which has a different number of modes for the two components, $\beta_2 - DPY$ is performing better than $\beta_1 - DPY$.

The evidence is in favour of the use of both DD and DPY priors under the specification (H1), with the exception of the cases where there is a strong heterogeneity across units, in terms of the number of clusters. In this case $\beta_2 - DPY$ prior should be preferred.

In order to corroborate our findings better, we study the sensitivity of our results to changes in the values of the parameter φ of the DPY process. This analysis also provides some hints for the specification of the prior distribution, using some known properties of the marginal DPY (and DD) processes such as the distribution of the number of clusters as a function of the parameters θ_1 , θ_2 and α and of the sample size T . Following a standard practice in Bayesian nonparametric, we do not estimate the hyperparameters θ_1 , θ_2 and α and set them instead to some fix values. For each parameter setting we compare the prior and posterior distributions of the number of clusters.

The prior distribution of the number of clusters K_T in the Pitman–Yor marginal processes is

$$P(K_T = k) = \frac{\alpha^{k-1} \Gamma(\theta/\alpha + k) \Gamma(\theta + 1)}{\Gamma(\theta/\alpha + 1) \Gamma(\theta + T)} S_\alpha(T, k)$$

where $S_\alpha(T, k)$ is a generalized Stirling number of the first kind, and $\Gamma(x)$ is the one-parameter gamma function (e.g., see Pitman (2006), Ch. 1 and 3). We use the following formula to evaluate the prior mean of the number of clusters for each marginal Pitman–Yor process:

$$\mathbb{E}(K_T) = \begin{cases} \sum_{j=1}^T \frac{\theta}{\theta + j - 1} & \text{if } \alpha = 0 \\ \frac{\Gamma(\theta + \alpha + T) \Gamma(\theta + 1)}{\alpha \Gamma(\theta + \alpha) \Gamma(\theta + T)} - \frac{\theta}{\alpha} & \text{if } \alpha \neq 0 \end{cases}$$

with $\theta = \theta_1 + \theta_2$ for both marginal DPY processes when using the Construction (H1) and $\theta = \theta_1 + \theta_2$ and $\theta = \theta_1$ for the first and second marginal DPY processes respectively, when using the Construction H2. These formulas can be used to set the parameters of the DPY priors, instead of including them in the inference procedure.

In order to obtain the posterior distribution of the number of clusters, we apply our MCMC algorithm for the non-parametric repeated measurement model. As a reference for the hyperparameter settings we consider the estimates obtained by applying our MCMC approximation procedure (see Table 4 in Appendix C). For the sake of brevity, in what follows we present the sensitivity results for the Mix4 dataset, under the prior specification $\beta_2 - DPY(\varphi, G_0)$.

Note that α ranges in $[0, 1]$ and K_T in $[1, T]$, T being the number of observations. Moreover, θ_1 and θ_2 have a (proper) gamma prior with finite (prior) moments of any order. The boundedness of α and K_T , and the prior assumption for θ_1 and θ_2 guarantee the existence of the posterior moments reported in Tables 4–5, Appendix C, and Table 3, Section 6.2.2.

In a first set of experiments we consider $\theta_1 = 0.09$ and $\theta_2 = 1.21$ and let α vary in the set $\{0.1, 0.2, 0.3\}$. For each parameter setting we compare the original posterior (dashed lines in Fig. 4) with the prior (solid lines in the same figure) and posterior (dotted lines in the same figure) distributions of the number of clusters for each marginal DPY process. The prior distribution is more dispersed about the mean than the posterior (see prior and posterior means and standard deviations in Table 5, panel (a)), and in some experiments the prior expected number of clusters is smaller than the true. Nevertheless, the posterior distribution is quite robust to prior specification in all our experiments. The posterior expected number of clusters is approximately 7 and 3 for the first and second components respectively. As a side remark, we note that an increase in the value of the parameter α usually leads to an increase of the computing time. This is due to the higher expected prior number of clusters and to the higher prior dispersion about the central value of the mixture components. For example, 10,000 MCMC iterations took about 15 min for $\alpha = 0.01$ and 50 min for $\alpha = 0.6$, on a 4 cores machine with a Intel i5 M540 2.53 GHz CPU, in MatLab.

In two other sets of experiments, we analyse the sensitivity of the posterior number of clusters with respect to θ_1 and θ_2 , when setting $\alpha = 0.02$. The results, reported in Appendix C, show that the prior dispersion and posterior number of clusters are not strongly affected by the values of the hyperparameters θ_1 and θ_2 .

Finally, we study the posterior sensitivity with respect to the hyper-parameters of the hierarchical base measure. We consider the parameter s_0^2 , which is the precision of the common mean parameter and which drives the prior dependence between the components of the DPY process. For $s_0^2 \in \{0.1, 1, 10\}$ we find that the posterior expected number of clusters is 6.92, 7.23 and 7.51 for the first component and 2.92, 3.23 and 3.47 for the second component. We conclude that the posterior distribution of the number of clusters (see Fig. 5) is not strongly affected to this hyperparameter.

6.2. $\beta_2 - DPY(\varphi, H_0)$ mixtures of vector autoregressive processes

6.2.1. Estimation results

In business cycle modelling great advances have been made by allowing for separate parameter values in periods (called regimes) of recession and expansion. The seminal paper of Hamilton (1989) proposes to use a dynamic mixture model (regime-switching model) with two components for capturing clustering of observations during the recession and expansion phases in a business cycle. This simple model has been successfully extended in many directions (e.g., see Krolzig (2000)) and the issue of estimating the number of regimes has been considered in various papers following a parametric approach. Conversely, in this paper, we apply a non-parametric approach to the estimation of the number of regimes or structural breaks. We extend their approach to a multivariate set-up and propose a joint estimation procedure for the number of regimes or breaks in multiple time series. We assume a $\beta_2 - DPY(\varphi, H_0)$ prior for the parameters of a two-country panel VAR and apply the resulting non-parametric model to two well studied business cycles: the United States (US) and the European Union (EU). Specifically it is interesting to verify whether the strong contraction in 2009 calls for the use of a higher number of regimes than three or four to achieve a better modelling of the cycles.

We consider seasonally and working day adjusted industrial production indexes (IPI), at a monthly frequency from the time of April 1971 to January 2011, for the US and the EU (see first row in Fig. 6). We take the quarterly growth rate of US and EU IPI, $\{Y_{1,1t}\}_{t=1}^T$ and $\{Y_{1,2t}\}_{t=1}^T$ respectively, (see the second row in Fig. 6). To achieve a better modelling of the business cycle we consider

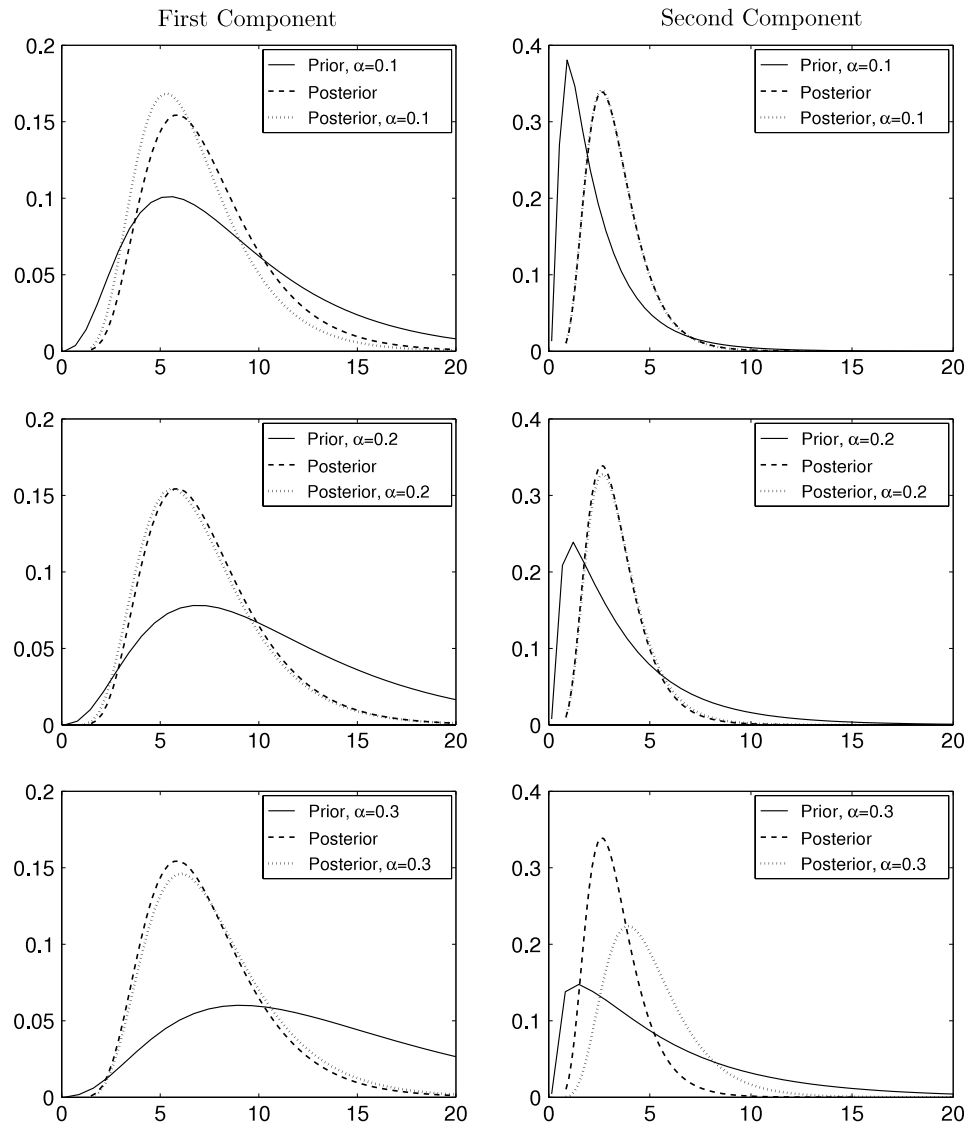


Fig. 4. Dotted lines: posterior distributions of the number of clusters, K_T for different values of α (different rows). Dashed lines: original posterior distribution with estimated φ . Solid lines: prior distribution for $\theta_1 = 0.09$, $\theta_2 = 1.21$, and $T = 100$.

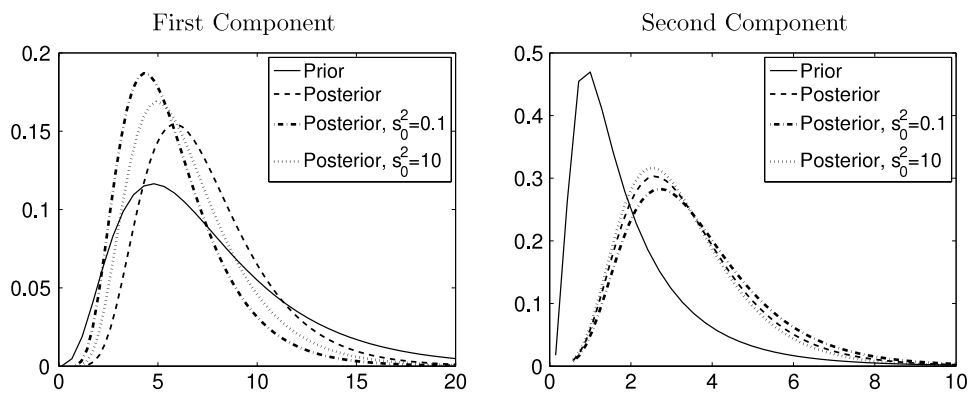


Fig. 5. Dotted and dashed-dotted lines: posterior distributions of the number of clusters for different values of s_0^2 . Dashed lines: original posterior distributions with estimated φ . Solid lines: prior distributions for $\theta_1 = 0.09$, $\theta_2 = 1.21$, $\alpha = 0.02$ and $T = 100$.

the term spread (TS), that is the difference between the 3-month and the 10-year interest rate for the US and the EU, $\{Y_{2,1t}\}_{t=1}^T$ and $\{Y_{2,2t}\}_{t=1}^T$ respectively, (see third row in Fig. 6).

As a preliminary analysis of the IPI data we apply the Bry and Boschan (1971) (BB) rule for the in-sample identification of the

downward and upward turns of a time series. This non-parametric technique is now standard in the literature (e.g., see Canova and Ciccarelli (2004) and Billio et al. (2012)) and in the practice of business cycle analysis. It is currently used by the NBER institute for extracting the reference cycle for the US. The BB rule identifies

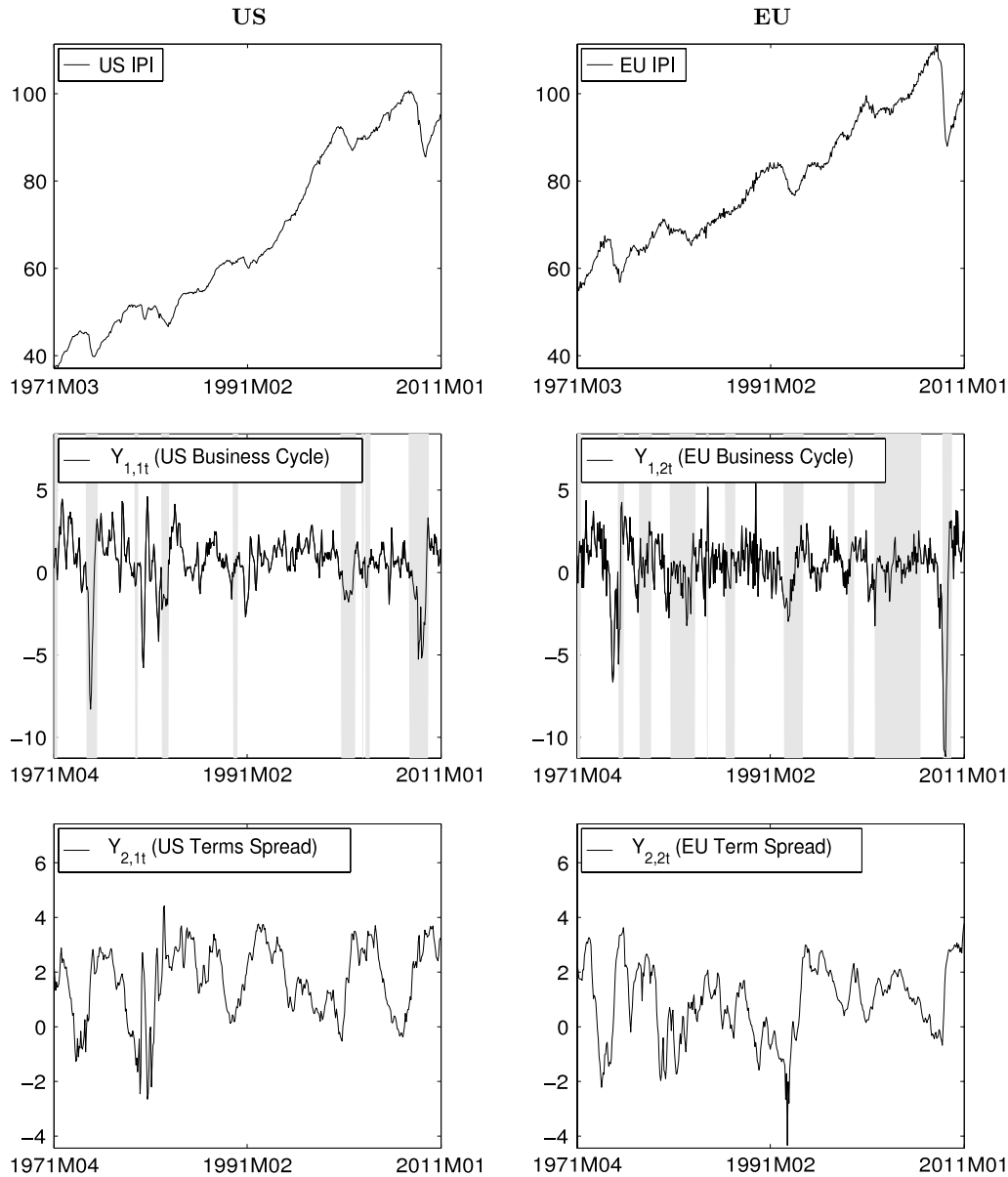


Fig. 6. First row: Industrial Production Index (IPI) for the US and the EU at a monthly frequency for the period: March 1971–January 2011. Second row: logarithmic quarterly changes in the US IPI ($Y_{1,1t}$) and the EU IPI ($Y_{1,2t}$) variables and NBER official recession phases (vertical grey bars). Third row: 10-year and 3-month interest rate spread (term spread) for the US ($Y_{2,1t}$) and the EU ($Y_{2,2t}$).

a downward turn (or peak) at time t if $x_{t-K} < x_t, \dots, x_{t-1} < x_t$ and $x_t > x_{t+1}, \dots, x_t > x_{t+K}$ and an upward turn (or trough) at time t if $x_{t-K} > x_t, \dots, x_{t-1} > x_t$ and $x_t < x_{t+1}, \dots, x_t < x_{t+K}$, where K usually takes values 3, 4 or 5. Similarly, we define a non-downward turn at time t if $x_{t-K} < x_t, \dots, x_{t-1} < x_t$ and $x_t < x_{t+1}, \dots, x_t < x_{t+K}$ and a non-upward turn at time t if $x_{t-K} > x_t, \dots, x_{t-1} > x_t$ and $x_t > x_{t+1}, \dots, x_t > x_{t+K}$. Note that at time t , the quantities $x_{t+1}, \dots, x_{t+K}$ are not known, thus the BB cannot be used for forecasting purposes. If we apply the BB rule in-sample, with $K = 3$, then we obtain the vertical bars in the second row of Fig. 6. The bars show the time periods from an upturn to a downturn (recession phases) of the US and the EU economies. Following the BB rule, one can identify, in both series, many recession periods, and in particular the 2009 recession. This simple procedure cannot be used out-of-sample to predict the turning points of the cycle and this calls for the use of suitable forecasting models, which are able to account for the relevant features of these series, such as skewness, heavy tails and multimodality in

the unconditional distribution (see the histograms in Fig. 7). These departures from the normality (see solid lines in Fig. 7) may be due to the presence of regimes, breaks or outliers in the series, which are usually modelled with shifts in the parameters of a linear model (see Krolzig (2000)). Thus, in our Bayesian non-parametric panel VAR model, we consider shifts in intercepts, autoregressive coefficients and covariance matrices. We assume cross-country conditional independence, as described in Section 2.2.2, and the hierarchical base measure given in Section 4.2 with $p = 4$. See Hamilton (1989) and Krolzig (2000) for a discussion on lag selection. In summary, the first stage equations of our hierarchical Bayesian model are

$$Y_{it} = (I_2 \otimes X_t') \phi_{it}^* + \varepsilon_{it} \quad (23)$$

for $i = 1, 2$ and $t = 1, \dots, T$, where $Y_{it} = (Y_{1,it}, Y_{2,it})'$, $X_t = (1, Y'_{1,t-1}, Y'_{2,t-1}, \dots, Y'_{1,t-p}, Y'_{2,t-p})'$, $\varepsilon_{it} = (\varepsilon_{1,it}, \varepsilon_{2,it})'$ with $\varepsilon_{it} \sim \mathcal{N}_2(0, \Sigma_{it}^*)$. As regard to the second and third stages of hierarchy,

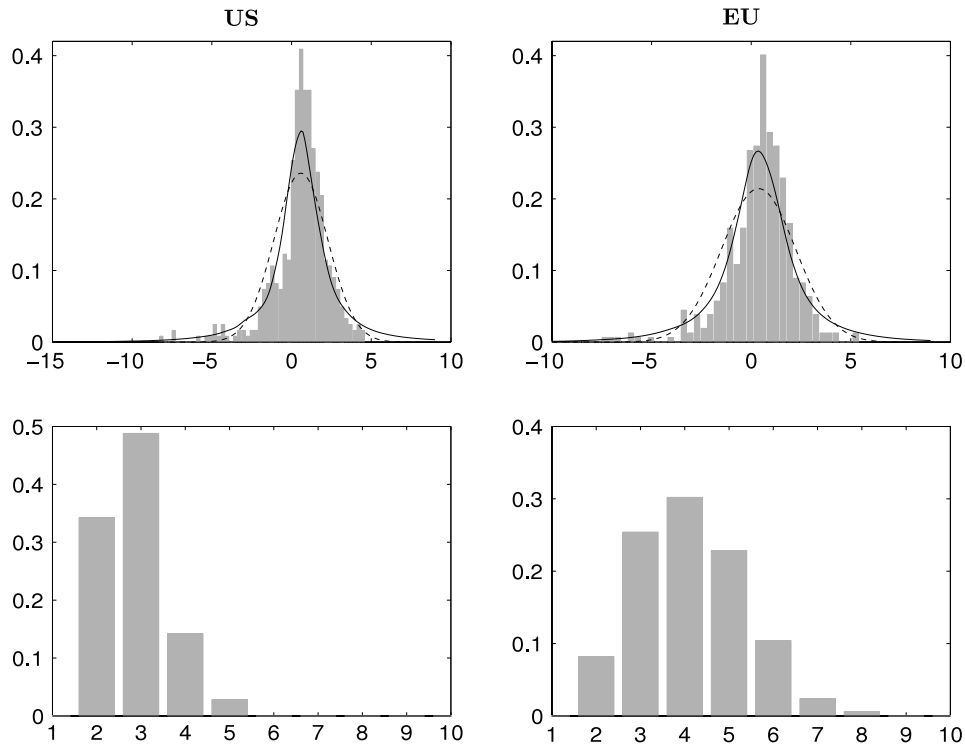


Fig. 7. First row: IPI log-changes (histogram), predictive distribution (solid line) and best normal (dashed line) for the US (left column) and the EU (right column). Second row: posterior distribution of the number of clusters for the US (left column) and the EU (right column).

the equations are

$$\begin{aligned}\phi_{it}^* &= \tilde{\phi}_{0D_{it}} + \tilde{\phi}_{iD_{it}} \\ \Sigma_{it}^* &= \tilde{\sigma}_{0D_{it}} \tilde{\Sigma}_{iD_{it}}\end{aligned}\quad (24)$$

$i = 1, 2$, where $\tilde{\phi}_{0k}$ is a common factor to all countries, $\tilde{\phi}_{ik}$ is a country-specific factor and $D_{it} \in \{1, \dots, D_i^*\}$ are the unit-specific allocation variables, which are generated by the Pitman–Yor model

$$\begin{aligned}(\phi_{ik}^*, \Sigma_{ik}^{*-1}) &\stackrel{i.i.d.}{\sim} G_i \quad i = 1, 2 \\ (G_1, G_2) &\sim \beta_2 - \text{DPY}(\tilde{\varphi}, G_0) \\ \tilde{\varphi} &\sim \mathcal{G}(\zeta_{11}, \zeta_{21}) \mathcal{G}(\zeta_{12}, \zeta_{22})\end{aligned}\quad (25)$$

with base measure G_0 given by the following hierarchical structure

$$\begin{aligned}(\tilde{\phi}_{ik}, \tilde{\Sigma}_{ik}^{-1}) &\sim \mathcal{N}(0, \tilde{\gamma}_{ik}) \mathcal{W}_2(\lambda, \Lambda) \quad i = 1, 2 \\ (\tilde{\phi}_{0k}, \tilde{\sigma}_{0k}^{-1}) &\sim \mathcal{N}(0, \gamma_0) \mathcal{G}(\varepsilon/2, \varepsilon/2),\end{aligned}\quad (26)$$

where $m = 2(4p + 1)$. In the same spirit of Chib and Greenberg (1995), we further assume that the hyperparameters $\lambda, \Lambda, \gamma_0$ and ε are known and that $\tilde{\gamma}_{ik} = \tilde{\tau}_k^2 I_m, i = 1, 2$, are random with

$$\tilde{\tau}_k^2 \sim \mathcal{IG}(\nu_0/2, \nu_0/2). \quad (27)$$

The Gibbs sampling implementation for the model in Eqs. (23)–(27) is given in Appendix B.3. The charts in the first row of Fig. 7 show the predictive distributions (solid lines) generated by the non-parametric approach conditioning on all values of $Y_t = (Y'_{1t}, Y'_{2t})', t = 1, \dots, T$, and the best normal fits (dashed lines) for the empirical distributions of the two series.

From a comparison with the empirical distribution, we note that the non-parametric approach, as opposed to the normal model, is able to capture skewness and excess of kurtosis in the data. The results from our model are in line with the practice of using time-varying parameter models (e.g., Markov-switching models) to capture asymmetry and non-linearity in both the US and the EU business cycle. The main results of our non-parametric

approach can be summarized through the implied data clustering and the posterior predictive densities.

As regard to the data clustering, the posterior distribution of the number of clusters is given in the second row of Fig. 7. The location of the posterior mode of the histograms allows us to conclude that following our non-parametric approach there is evidence in favour of three clusters for the US cycle and at least of four clusters for the EU cycle. The result for the US data is coherent with the results available in the literature where three-regime Markov-switching models (see for example Krolzig (2000)) are usually considered. The result for the EU cycle is, in a certain way, coherent with the output of parametric studies which suggest to consider at least three regimes. Nevertheless, the effects of the 2009 recession on the past empirical findings is an open issue and a matter of research. The result from our non-parametric approach is interesting because it suggests a substantial evidence in favour of a larger number of mixture components, at least four, for the EU than the one used for the US (see Fig. 7). The identification of the mixture components relies upon the implied data clustering. In order to estimate the implied data clustering of our DPY mixture model, we apply the least square clustering method proposed originally in Dahl (2006). The method has been successfully used in many applications (e.g., see Rodriguez et al. (2008)) and is based on the posterior pairwise probabilities of joint classification $P\{D_{is} = D_{it}|Y\}$. To estimate this matrix, one can use the following pairwise probability matrix:

$$P_{i,st} = \frac{1}{M} \sum_{l=1}^M \delta_{D_{is}^l} (D_{it}^l)$$

that is estimated by using every pair of allocation variables D_{is}^l and $D_{it}^l, s, t = 1, \dots, T$ and $l = 1, \dots, M$, where M is the number of MCMC iterations.

Fig. 8 shows the pairwise posterior probabilities $P_{i,st}, i = 1, 2$, of the US and the EU data for $s, t \in \{1, \dots, T\}$. The first row (second row) shows the posterior probabilities that two observations of

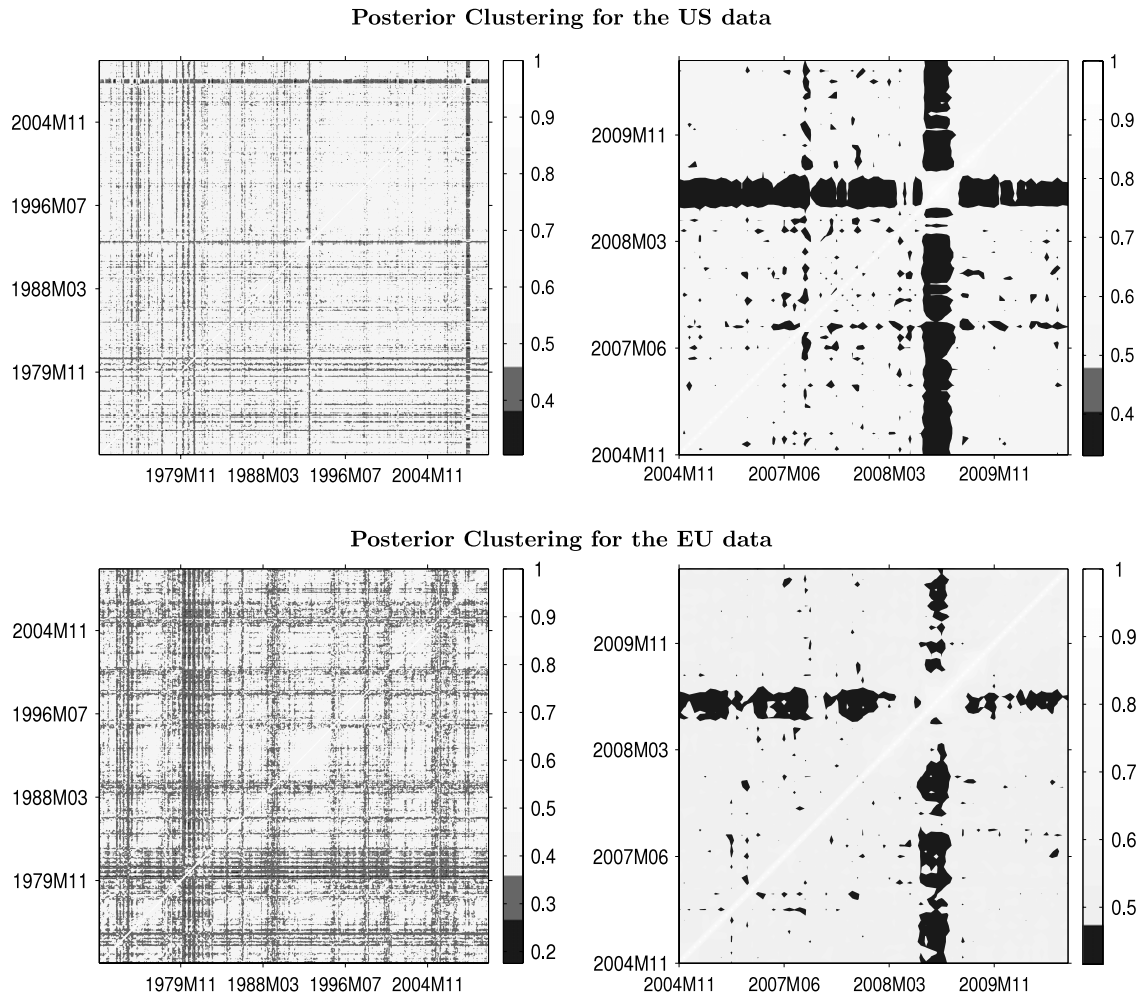


Fig. 8. Pairwise posterior probabilities for the clustering of the US data $P_{1,st}$ and the EU data $P_{2,st}$ for $s, t \in \{1, \dots, T\}$.

the US cycle (EU cycle) belong to the same cluster. In the first column, one can detect the presence of vertical and horizontal dark grey bands. They correspond to observations that do not cluster frequently together with other observations. A similar remark is true for the light grey areas. In the second column of Fig. 8, one can see the different behaviour of the clustering for the US and the EU during the 2009 crisis. As the observations about January 2009 group together in a cluster which excludes other observations of the sample, we can identify this component of the mixture as the one associated to the 2009 recession. The same interpretation of the implied data clustering and the identification of the components as the fourth regime for the 2009 recession can be also achieved through the least square marginal clustering, that is the clustering $D_i^l = (D_{i1}^l, \dots, D_{iT}^l)$ sampled at the l_i -th iteration which minimizes the sum of squared deviations from the pairwise posterior probability:

$$l_i = \arg \min_{l \in \{1, \dots, M\}} \sum_{t=1}^T \sum_{s=1}^T \left(\delta_{D_{it}^l} (D_{it}^l) - P_{i,st} \right)^2.$$

As regard to the predictive densities generated by the DPY model, we observe that they have long left tails (solid lines in Fig. 7), fatter than the tails of the best normal (dashed lines in the same figure). Thanks to these features the DPY model is suitable for describing and predicting these data. One of the features of a Bayesian nonparametric model is its flexibility in fitting the marginal posterior predictive densities of the data and this flexibility can be exploited when forecasting the turning

points of the business cycle. We show how to use the predictive densities of our $\beta_2 - \text{DPY}$ model in combination with the BB rule, to produce out-of-sample forecast of the upward and downward turns of the cycle. More specifically, let y_t be a variable of interest (i.e., the US and EU industrial production growth rates, $Y_{1,1t}$ and $Y_{1,2t}$ respectively), then we set $K = 1$ as in Canova and Ciccarelli (2004) and evaluate the following downward and upward turn probabilities

$$P_t^{(D)} = \int_{-\infty}^{+\infty} \mathbb{I}_{[y_{t-1}, +\infty)}(y_t) \mathbb{I}_{[y_{t+1}, +\infty)}(y_t) f(Y_{t+1} | Y_{1:t}) dY_{t+1}$$

$$P_t^{(U)} = \int_{-\infty}^{+\infty} \mathbb{I}_{(-\infty, y_{t-1}]}(y_t) \mathbb{I}_{(-\infty, y_{t+1}]}(y_t) f(Y_{t+1} | Y_{1:t}) dY_{t+1}$$

where $f(Y_{t+1} | Y_{1:t})$ is the predictive density of the $\beta_2 - \text{DPY}(\varphi, G_0)$ process. These integrals can be naturally approximated using the MCMC sample from the predictive density. Under the assumption of the symmetric loss function, we predict a peak at time t if $P_t^D > 0.5$ and a trough if $P_t^U > 0.5$. The resulting cycle is: $z_t = z_{t-1} \mathbb{I}_{[0, 0.5]}(P_t^D) + (1 - z_{t-1}) \mathbb{I}_{(0.5, 1]}(P_t^U)$, that is equal to 1 in the expansion phases and 0 in the recession phases. We assume z_0 is given. We stress that, in the proposed method, the prediction of a turning point at time t needs the predictive density for the variable of interest at time $t + 1$, i.e. y_{t+1} .

Fig. 9 shows a typical predictive density approximation of the US (left) and the EU (right) growth rate from our model during expansion (e.g., observation $t = 430$, 1st July 2007) and recession (e.g., observation $t = 450$, 1st March 2009) periods, where

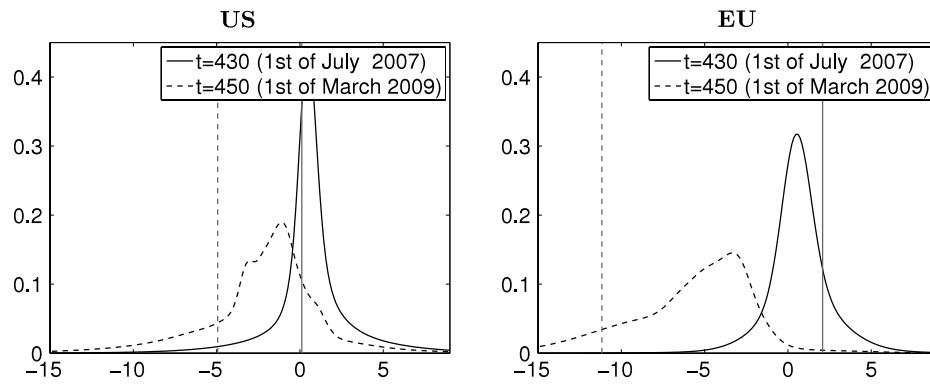


Fig. 9. Approximated predictive densities of the US (left) and the EU (right) IPI growth rate, $Y_{1,t}$ and $Y_{1,2t}$ (vertical lines), during expansion ($t = 430$) and recession ($t = 450$) periods.

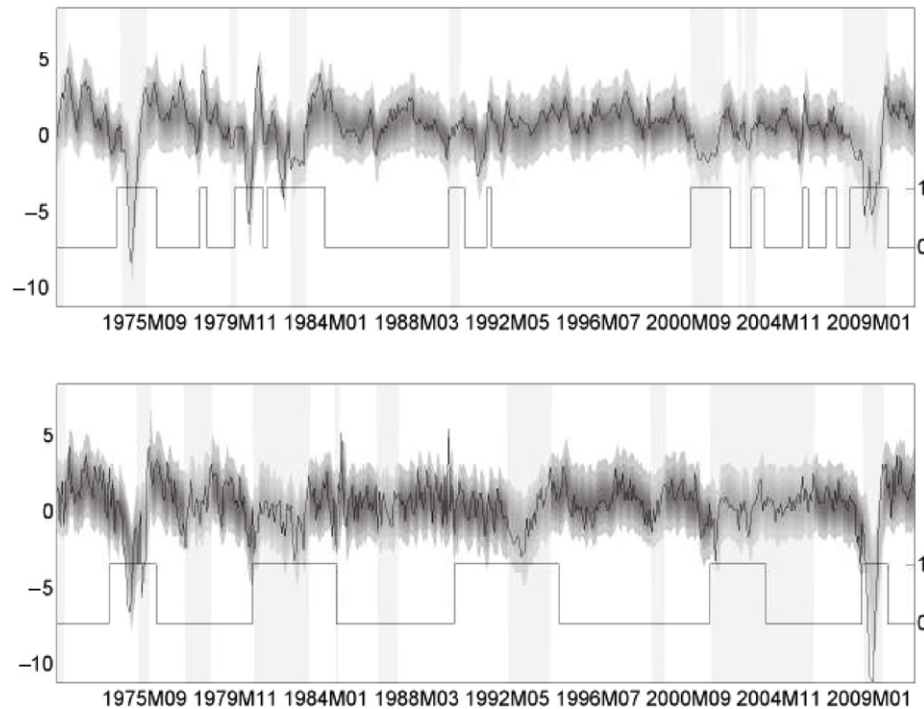


Fig. 10. Predictive results for US (top) and EU (bottom). In each plot: IPI growth rates (black lines, left axis); heatmap (grey areas, left axis) of the 95% high probability density region of the predictive density functions (darker colours represent higher density values) evaluated at each time point, for $t = 1, \dots, T$, at the values of the predictors $Y_{t-1}, \dots, Y_{t-p}$, for $i = 1, 2$; reference cycle (vertical grey bars) and predicted reference cycle (stepwise line, right axis).

expansion and recession have been identified by applying the BB rule for business cycle classification. The posterior predictive densities during expansion phases exhibit skewness with right tails slightly fatter than left tails. During the recession period the predictive distributions exhibit multimodality with at least two modes in the negative half. The main differences between the two cycles are that the EU cycle exhibits more pronounced modes in the negative half, say around -11 and -5 , and that the volatility of the mixture components is higher.

Fig. 10 shows the one-step-ahead posterior predictive densities for $Y_{1,it}$, $i = 1, 2$, estimated sequentially over time, $t = 50, \dots, 474$, and evaluated at the current values of the explanatory variables $Y_{it-1}, \dots, Y_{it-p}$ ($i = 1, 2$). In the same plot, the grey area represents the heatmap sequence of the 95% high probability density region of the predictive densities (darker colours represent higher density values). These densities have been used to forecast the peaks and troughs of the cycles (stepwise line) following the predictive BB rule given above. The predicted business cycle is close to the reference cycle generated in-sample using a BB rule with $K = 3$ (see grey vertical bars). In order to measure the

similarity between the two cycles, we follow a standard practice in business cycle analysis and evaluate the concordance statistic proposed by [Harding and Pagan \(2002\)](#). This statistic is a measure of the proportion of time during which the two series, e.g. the reference cycle and the predicted cycle, are in the same state. This measure ranges between 0 and 1, with 0 representing perfectly counter-cyclical switches, and 1 perfectly synchronous shifts. For two regimes described by random walks, the measure will be 0.5 in the limit. The concordance statistics are 0.7594 and 0.8379 for the EU and the US cycle respectively, which provides evidence of a good predictive ability of our models. A more detailed analysis of the features, such as average phase duration and average phase amplitude (see [Harding and Pagan \(2002\)](#)), of the cycle generated by our non-parametric predictive model and a comparison with other methods are not the main goal of this paper and we leave this for further research.

6.2.2. Prior selection and sensitivity analysis

In order to assess the predictive ability of the proposed β_2 – DPY(φ, H_0) two-country panel VAR model, we compare it with

Table 2

In each row: the predictive considered (first column), the optimal pooled log-predictive score function (second column) and the optimal weights (remaining columns) for independent Dirichlet (w_1), $\beta_1 - DD(w_2)$, $\beta_2 - DD(w_3)$, $\beta_1 - DPY(w_4)$ and $\beta_2 - DPY(w_5)$, for the periods 1975M08–1996M05 (panel (a)) and 1975M08–2010M12 (panel (b)).

Joint predictives	Log-score	w_1	w_2	w_3	w_4	w_5
Panel (a)						
US–EU IPI	–169.72	0.0000	0.5263	0.0000	0.3619	0.1118
US–EU IPI and TS	–350.91	0.0000	0.3203	0.0000	0.6797	0.0000
Panel (b)						
US–EU IPI	–453.44	0.0004	0.1050	0.0000	0.3302	0.5644
US–EU IPI and TS	–912.52	0.0000	0.2512	0.0000	0.6275	0.1213

other non-parametric prior specifications: two unit-specific and independent Dirichlet process priors, the dependent Dirichlet process priors, $\beta_i - DD(\varphi, H_0)$, $i = 1, 2$ and the dependent Pitman–Yor process prior, $\beta_1 - DPY(\varphi, H_0)$. We apply, as in the previous section, the model pooling approach of Geweke and Amisano (2010) based on the log predictive score. We consider two alternative sets of predictive densities: the joint US–EU IPI density $f(Y_{1,1t}, Y_{1,2t} | Y_1, \dots, Y_{t-1}, M_j)$, $j = 1, \dots, J$, and the joint US–EU and IPI–TS predictive density $f(Y_t | Y_1, \dots, Y_{t-1}, M_j)$, $j = 1, \dots, J$, where $Y_t = (Y'_{1t}, Y'_{2t})'$. In the previous predictive densities, M_j indicates the following models: independent Dirichlet ($j = 1$), $\beta_1 - DD(\varphi, G_0)$ ($j = 2$), $\beta_2 - DD(\varphi, G_0)$ ($j = 3$), $\beta_1 - DPY(\varphi, G_0)$ ($j = 4$) and $\beta_2 - DPY(\varphi, G_0)$ ($j = 5$). We define the combined predictive and the log pooled predictive score function as in the previous section. We approximated the model-specific predictive densities by using the Gibbs sampling scheme given in Appendix B. We evaluate the log pooled predictive score function on two different subsets of the data with the aim to check the effect of the 2009 crisis on model selection. More specifically, for the first sub-sample we consider observations from August 1975 to May 1996 and set $\tau_1 = 50$ and $\tau_2 = 200$ in the pooled score function, while for the second sub-sample we consider the time period January 1979–December 2010 (i.e., we set $\tau_1 = 50$ and $\tau_2 = 474$ in the pooled score function). We follow the same procedure of the previous section to obtain a sequence of predictive densities and an optimal combination weight for each model. The results in Table 2 show that for the two subsamples the DD and DPY models have the better predictive ability in terms of both US–EU IPI and US–EU IPI and TS joint predictive densities. As regard to the choice of the model within the class of vector of dependent Dirichlet, we found that in the first subsample $\beta_1 - DD(\varphi, H_0)$ allows for a better modelling of the US–EU IPI joint predictive (see Panel (a) of Table 2) while on the whole sample $\beta_2 - DD(\varphi, H_0)$ has better predictive ability. Finally, we note that the $\beta_1 - DPY(\varphi, H_0)$ VAR for the joint prediction of IPI and TS, for US and EU, has the highest combination weight on the whole set of data.

The evidence from this study, is in favour of the use of both DD and DPY priors in VAR models to capture multiple breaks or regimes in the parameters of the model. From our experiments, the DPY is the prior specification for a multi-unit panel VAR that produces the best results in terms of prediction. The assumption (H1) should be preferred to (H2) when no prior information is available on the number of clusters for the different units of the panel. In our application both assumptions (H1) and (H2) lead to models with good predictive abilities. In the analysis of the EU and US IPI growth rates one can expect a priori, from the existing studies, that the number of clusters differs across the two countries and that the number of clusters for the EU is larger. Thus, the elicitation of a $\beta_2 - DPY$ prior can easily motivated in this example. On the whole sample, the $\beta_2 - DPY$ prior produces better prediction of the IPI growth rates than a $\beta_1 - DPY$ prior.

Also, we carry out three sensitivity analyses to measure the robustness of the results of Section 6.2.1 regarding the selection of the hyper-parameters of the $\beta_2 - DPY$ process prior.

Table 3

Posterior and prior means ($\bar{k}_i = \mathbb{E}(K_i) | Y$), $k_i = \mathbb{E}(K_i)$ respectively) and standard deviations (in parenthesis) of the number of clusters for the first and second components of the $\beta_2 - DPY$ process for the IPI dataset. The sample size is $T = 474$ and the parameter setting is $\theta_1 = 0.1$, $\theta_2 = 0.7$, varying the value of α (panel (a)) and $\theta_1 = 0.4$, $\theta_2 = 0$ and $\alpha = 0.3$ (panel (b)).

α	$\bar{k}_1$	$\bar{k}_2$	k_1	k_2
Panel (a)				
0.04	3.80 (1.01)	2.51 (0.71)	6.53 (2.50)	2.04 (1.12)
0.1	4.34 (1.98)	3.21 (1.19)	11.30 (3.85)	2.76 (1.75)
0.2	5.99 (1.41)	3.79 (0.98)	12.023 (5.23)	4.84 (3.27)
0.3	5.77 (1.85)	3.84 (1.12)	16.08 (8.84)	7.94 (5.43)
Panel (b)				
0.3	9.23 (2.05)	4.21 (1.35)	13.09 (7.40)	13.09 (7.40)

First, we study the sensitivity of our results to changes in the values of stick-breaking concentration and dispersion parameters. We recall that, in the application to IPI data, the first component of the $\beta_2 - DPY(\varphi, G_0)$ process is used as a prior distribution in modelling the EU variables and the second component in modelling the US variables. Thus, the marginal prior distributions are a PY process of parameters $(\alpha, \theta_1 + \theta_2)$ for the EU and a PY process of parameters (α, θ_1) . The posterior means of θ_1 , θ_2 and α are 0.1013, 0.7140 and 0.0469, the standard deviations 0.0569, 0.3626 and 0.0393 and the credibility intervals [0.0135, 0.2132], [0.2225, 1.6407] and [0.0112, 0.1659]. In this analysis, we set $\theta_1 = 0.1$ and $\theta_2 = 0.7$ and focus on the effects of the parameter α on the posterior inference. Table 3 reports the prior and the posterior expected number of clusters and its standard deviation. Increasing α increases the posterior expected number of clusters and also its standard deviation (see also the shape of the posterior in panel (a) of Fig. 11). In all the experiments, the first component exhibits a posterior expected number of clusters which is much smaller than the prior one and larger than the one of the second component. The second component exhibits a number of clusters close to 3 or 4 under both prior assumptions of a smaller or larger number of clusters. This confirms the results we obtained in the original analysis, where all hyperparameters have been estimated on the data.

Secondly, we also consider the sensitivity regarding simultaneous changes in all the parameters α , θ_1 and θ_2 (see panel (b) of Fig. 11). More specifically we set $\theta_2 = 0$, which corresponds to the limiting case, discussed in Section 3.1, in which the Pitman–Yor marginal processes have a common stick-breaking structure. This assumption is far from what we found in the data, as the estimated θ_2 is 0.7140. Moreover we set $\theta_1 = 0.4$ and $\alpha = 0.3$ in order to have a large prior number of clusters. The prior expected number is about 13. Note that setting high values of θ_1 and α may lead to a substantial increase in the posterior expected number of clusters. In practice, we suggest either to include these parameters in the inference process or to use small values of θ_1 and θ_2 and large values

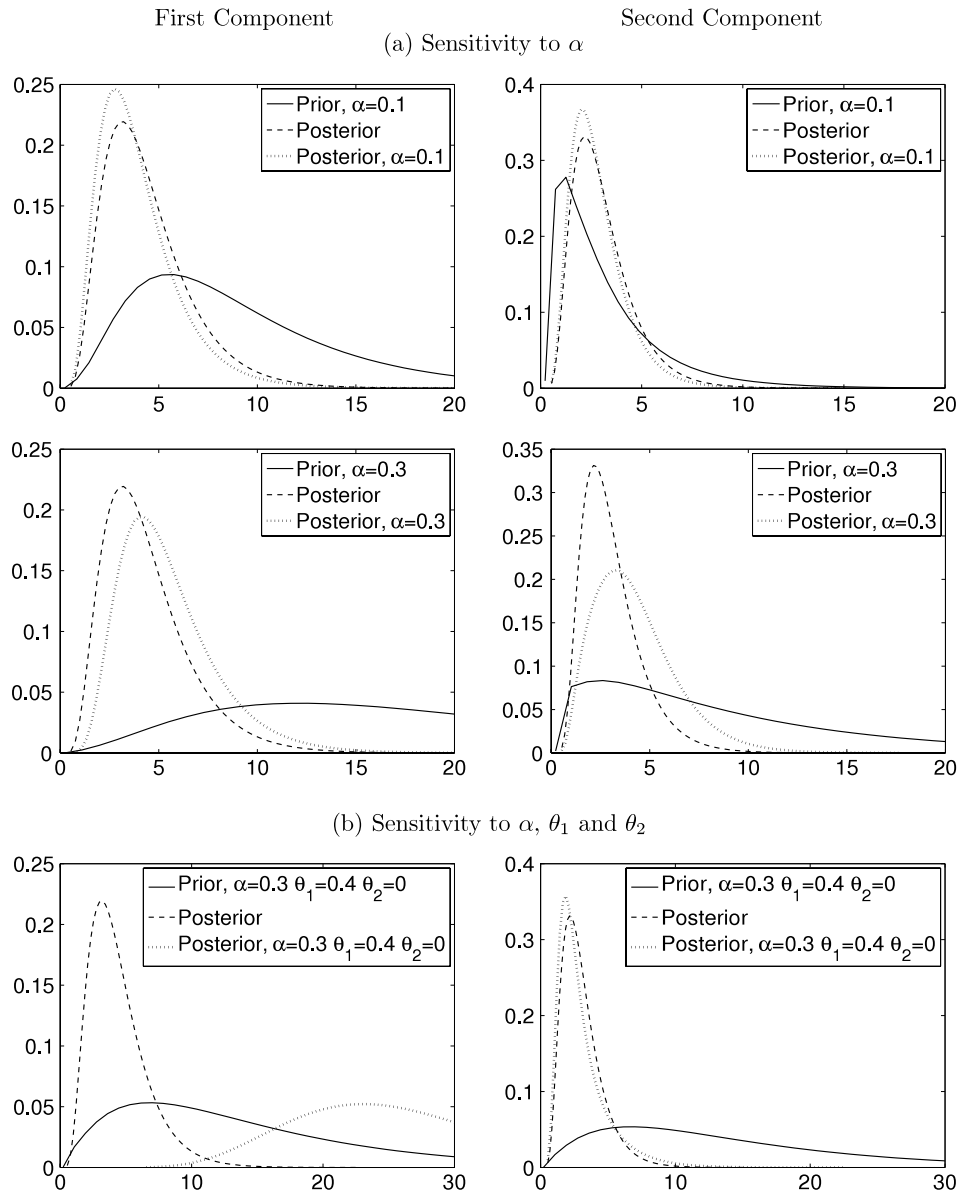


Fig. 11. Dotted lines: posterior distributions of the number of clusters, K_T , for different values of α (a) and for $\alpha = 0.3$, $\theta_1 = 0.4$, $\theta_2 = 0$ (b). Dashed lines: original posterior distributions with estimated φ . Solid lines: prior distributions for $\theta_1 = 0.1$, $\theta_2 = 0.7$ (a) and $\theta_1 = 0.4$, $\theta_2 = 0$ (b).

for α . However, in our experiments, we found that the posterior expected number of clusters of the first component is always higher than the one of the second component. Thus, one of the main findings of our study remains valid, i.e. a substantial evidence in favour of at least four components in the mixture for the EU data and a smaller number of components for the US data. This implies that dynamic mixture or Markov-switching models for EU data should account for a higher number of regimes than similar models for the US data.

Finally, we check the robustness of our results with respect to the specification of one of the hyperparameters of the hierarchical base measure. In the application the hyperparameter setting was as follows: $\lambda = 3$, $\Lambda = 10I_2$, $\gamma_0 = 0.001$, $\varepsilon = 10$ and $\nu_0 = 100$. We focus on the scale hyperparameter Λ of the component $\tilde{\Sigma}_{ik}^{-1}$ of the atoms, keep fixed the other hyperparameters and set $\theta_1 = 0.1$, $\theta_2 = 0.7$ and $\alpha = 0.04$. Fig. 12 shows how Λ affects the posterior number of clusters. Increasing the prior variance (or decreasing the prior precision) of the component-specific part of the atoms sensibly reduces the posterior expected number of clusters.

7. Conclusions

This paper proposes two classes of prior distributions, called beta-product dependent Pitman–Yor (DPY) processes, $\beta_1 - \text{DPY}(\varphi, G_0)$ and $\beta_2 - \text{DPY}(\varphi, G_0)$, with a hierarchical specification of the base measure, which allows for heterogeneous clustering effects and for information pooling across different groups of time series. The two classes of distributions extend the independent Dirichlet priors and have different features. The $\beta_1 - \text{DPY}(\varphi, G_0)$ should be used as a prior when either the clustering is expected to be similar along the different vector components or no prior information is available on the number of clusters for the different units. The $\beta_2 - \text{DPY}(\varphi, G_0)$, should be used instead when one expects that the clustering features differ across units. The dependent Pitman–Yor priors have the dependent-Dirichlet priors as special cases. The evidence in our simulation exercises and real data applications are in favour of both DPY and DD against the independent Dirichlet prior or a common Dirichlet prior. DPY should be preferred to DD when one expects a strong heterogeneity across units, in terms of clusters, outliers, breaks or regimes.

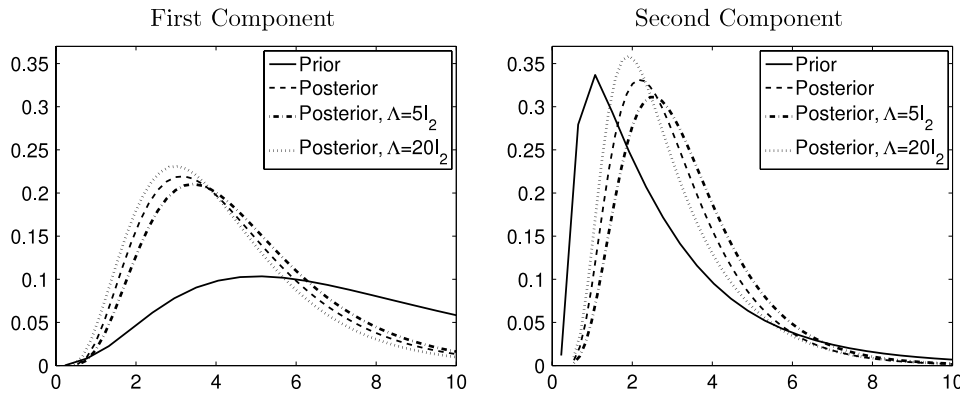


Fig. 12. Dotted and dashed–dotted lines: posterior distributions of the number of clusters, K_T , for different values of Λ . Dashed lines: original posterior distributions with estimated φ . Solid lines: prior distributions for $\alpha = 0.04$, $\theta_1 = 0.1$, $\theta_2 = 0.7$, and $T = 474$.

The proposed prior distributions are used to define new Bayesian non-parametric models for repeated measurement and multi-unit VAR models. We provide an efficient Monte Carlo Markov Chain algorithm for posterior computation, extending the exiting slice sampling algorithms for Bayesian nonparametric models. The effectiveness is assessed first in simulation exercises and then through an application to a multi-country macroeconomic dataset. Specifically, we consider the EU and US IPI growth rate series and estimate a two-country panel VAR with DPY priors. We found evidence of substantial difference in the clustering features of the EU and US series and of good predictive abilities of our non-parametric models for the turning points of the EU and the US business cycle. A sensitivity analysis shows that our main finding is not affected by changes in the prior hyperparameters.

We conclude the paper with the indication of some future research lines. Our approach can be extended by defining new nonparametric models in the context of repeated measurement and to large panel dataset. Thus the challenge is the use of parsimonious model specifications in combination with the DPY prior. Also a combination of our dependent-Dirichlet approach with the autoregressive Dirichlet process prior (e.g., see Griffin and Steel (2011) and Taddy (2010)) could be a possible extension which allows to capture time variations in the clustering effects. Also, the design of different inference procedures, such as particle filtering or learning (e.g., see Taddy (2010)), for on-line inference on our processes is left for future research.

Appendix A. Proofs

A.1. Proof of Proposition 3

First of all observe that

$$\begin{aligned} \mathbb{E}[G_i(A)G_j(B)] &= \sum_{h \geq 1, k \geq 1} \mathbb{E}[\mathbb{I}_A(\tilde{\psi}_{ik})\mathbb{I}_B(\tilde{\psi}_{jh})]\mathbb{E}[W_{ik}W_{jh}] \\ &= G_{0,i}(A)G_{0,j}(B) \sum_{h \geq 1, k \geq 1, h \neq k} \mathbb{E}[W_{ik}W_{jh}] \\ &\quad + G_{0,ij}(A \times B) \sum_{h \geq 1} \mathbb{E}[W_{ih}W_{jh}]. \end{aligned} \quad (28)$$

Now note that

$$\begin{aligned} \sum_{h \geq 1} \mathbb{E}[W_{ih}W_{jh}] &= \sum_{h \geq 1} \mathbb{E}\left[S_{1h}S_{2h} \prod_{m \leq h-1} (1 - S_{1m})(1 - S_{2m}) \right] \\ &= \frac{\mathbb{E}[S_{11}S_{21}]}{\mathbb{E}[S_{11}] + \mathbb{E}[S_{21}] - \mathbb{E}[S_{11}S_{21}]} \end{aligned} \quad (29)$$

and

$$\begin{aligned} &\sum_{h \geq 1, k \geq 1, h \neq k} \mathbb{E}[W_{ik}W_{jh}] \\ &= \sum_{h \neq k} \mathbb{E}\left[S_{1h}S_{2k} \prod_{m \leq h-1} (1 - S_{1m}) \prod_{l \leq k-1} (1 - S_{2l}) \right] \\ &= \frac{\mathbb{E}[S_{11}] + \mathbb{E}[S_{21}] - 2\mathbb{E}[S_{11}S_{21}]}{\mathbb{E}[S_{11}] + \mathbb{E}[S_{21}] - \mathbb{E}[S_{11}S_{21}]} \end{aligned} \quad (30)$$

Combining (28) with (29)–(30) it follows that

$$\begin{aligned} \mathbb{E}[G_i(A)G_j(B)] &= G_{0,ij}(A \times B) \frac{\mathbb{E}[S_{i1}S_{j1}]}{\mathbb{E}[S_{i1}] + \mathbb{E}[S_{j1}] - \mathbb{E}[S_{i1}S_{j1}]} \\ &\quad + G_{0i}(A)G_{0j}(B) \frac{\mathbb{E}[S_{i1}] + \mathbb{E}[S_{j1}] - 2\mathbb{E}[S_{i1}S_{j1}]}{\mathbb{E}[S_{i1}] + \mathbb{E}[S_{j1}] - \mathbb{E}[S_{i1}S_{j1}]} \end{aligned}$$

Since $\mathbb{E}[G_i(\cdot)] = G_{0i}(\cdot)$, $i = 1, \dots, r$, one gets

$$\begin{aligned} \text{Cov}[G_i(A), G_j(B)] &= \frac{\mathbb{E}[S_{i1}S_{j1}]}{\mathbb{E}[S_{i1}] + \mathbb{E}[S_{j1}] - \mathbb{E}[S_{i1}S_{j1}]} \\ &\quad \times [G_{0,ij}(A \times B) - G_{0i}(A)G_{0j}(B)]. \end{aligned} \quad (31)$$

In a similar way, for every $i = 1, \dots, r$, one has

$$\text{Var}[G_i(A)] = G_{0i}(A)(1 - G_{0i}(A)) \frac{\mathbb{E}[S_{i1}^2]}{2\mathbb{E}[S_{i1}] - \mathbb{E}[S_{i1}^2]}. \quad (32)$$

Hence,

$$\text{Cor}[G_i(A), G_j(A)] = C_{ij} \frac{G_{0ij}(A \times B) - G_{0i}(A)G_{0j}(B)}{\sqrt{G_{0i}(A)(1 - G_{0i}(A))G_{0j}(B)(1 - G_{0j}(B))}}$$

with

$$\begin{aligned} C_{ij} &= \frac{\mathbb{E}[S_{i1}S_{j1}]}{1 - \mathbb{E}[(1 - S_{i1})(1 - S_{j1})]} \\ &\quad \times \sqrt{\frac{(2\mathbb{E}[S_{i1}] - \mathbb{E}[S_{i1}^2])(2\mathbb{E}[S_{j1}] - \mathbb{E}[S_{j1}^2])}{\mathbb{E}[S_{j1}^2]\mathbb{E}[S_{i1}^2]}}. \end{aligned}$$

Assuming (14), one gets

$$\mathbb{E}(S_{i1}) = \frac{1}{1 + \theta_1 + \theta_2}, \quad \mathbb{E}(S_{i1}^2) = \frac{2}{(1 + \theta_1 + \theta_2)(2 + \theta_1 + \theta_2)}$$

$$\mathbb{E}(S_{i1}S_{j1}) = \frac{2(1 + \theta_1)}{(2 + \theta_1)(1 + \theta_1 + \theta_2)^2}$$

for every $i, j = 1, \dots, r$ and then,

$$C_{ij} = \frac{(1 + \theta_1 + \theta_2)(1 + \theta_1)}{(1 + \theta_1)(1 + \theta_1 + \theta_2) + \theta_2}.$$

Let us assume now (15). For the sake of simplicity write V_k in place of V_{k1} . Recall that $V_0 \sim \text{Beta}(1, \theta_1)$ and, for $1 \leq k \leq r-1$, $V_k \sim \text{Beta}(1 + \theta_1 + \dots + \theta_k, \theta_{k+1})$. Let $1 \leq i < j \leq r$. Since $S_{i1} = V_0 V_1 \dots V_{r-i}$, one gets

$$S_{i1} S_{j1} = V_0^2 V_1^2 \dots V_{r-j}^2 V_{r-j+1} \dots V_{r-i}.$$

After some computations, using also the fact that the V_j s are independent, one obtains

$$\mathbb{E}[S_{i1} S_{j1}] = \frac{2}{(2 + \theta_1 + \dots + \theta_{r-j+1})(1 + \theta_1 + \dots + \theta_{r-i+1})}$$

and

$$\mathbb{E}[S_{i1}] = \frac{1}{1 + \theta_1 + \dots + \theta_{r-i+1}},$$

$$\mathbb{E}[S_{i1}^2] = \frac{2}{(1 + \theta_1 + \dots + \theta_{r-i+1})(2 + \theta_1 + \dots + \theta_{r-i+1})}.$$

At this stage, simple algebra gives

$$\sqrt{\left(\frac{2\mathbb{E}[S_{i1}]}{\mathbb{E}[S_{i1}^2]} - 1\right)\left(\frac{2\mathbb{E}[S_{j1}]}{\mathbb{E}[S_{j1}^2]} - 1\right)} = \sqrt{(1 + \theta_1 + \dots + \theta_{r-i+1})(1 + \theta_1 + \dots + \theta_{r-j+1})}$$

and

$$\frac{\mathbb{E}[S_{i1} S_{j1}]}{\mathbb{E}[S_{i1}] + \mathbb{E}[S_{j1}] - \mathbb{E}[S_{i1} S_{j1}]} = \frac{2(1 + \theta_1 + \dots + \theta_{r-j+1})}{2(1 + \theta_1 + \dots + \theta_{r-j+1})^2 + (2 + \theta_1 + \dots + \theta_{r-j+1})(\theta_{r-j+2} + \dots + \theta_{r-i+1})}.$$

That is

$$C_{i,j} = \frac{2\sqrt{(1 + \theta_1 + \dots + \theta_{r-i+1})(1 + \theta_1 + \dots + \theta_{r-j+1})}^{\frac{3}{2}}}{2(1 + \theta_1 + \dots + \theta_{r-j+1})^2 + (2 + \theta_1 + \dots + \theta_{r-j+1})(\theta_{r-j+2} + \dots + \theta_{r-i+1})}.$$

Formula (17) can be proved in an analogous way.

A.2. The joint distribution of $[V, \tilde{\psi}, U, D, Y, \tilde{\varphi}]$ and related formulas

The joint distribution of $[V, \tilde{\psi}, U, D, Y, \tilde{\varphi}]$ is

$$\begin{aligned} P\{V \in dv, \tilde{\psi} \in d\psi, Y \in dy, U \in du, D = d, \tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha)\} \\ = \left[\prod_{i=1,2} \prod_{t=1}^{T_i} \mathbb{I}\{u_{it} < w_{i,d_{it}}\} \mathcal{K}_t(y_{i,t} | \psi_{d_{it}}, z_t) \right] dy du \\ \times_{k \geq 1} [P\{V_k \in dv_k | \tilde{\varphi} = (\theta_1, \theta_2, \alpha)\} G_0(d\psi_k)] \\ \times P\{\tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha)\} \end{aligned} \quad (33)$$

where $w_{i,k} = v_{0k} v_{ik} \prod_{t < k} (1 - v_{0t} v_{it})$, with the convention that $v_{2k} = 1$, for every k , under (H2) and $z_t = [y_{js} : j = 1, 2; s = 1, \dots, t-1]$. From (33) one gets

$$\begin{aligned} P\{V \in dv, \tilde{\varphi} \in (d\theta_1, d\theta_2, l) | Y, \tilde{\psi}, D\} \\ \propto \left[\prod_{i=1,2} \prod_{t=1}^{T_i} w_{i,D_{it}} \right] \times_{t \geq 1} P\{V_t \in dv_t | \tilde{\varphi} = (\theta_1, \theta_2, \alpha)\} \\ \times P\{\tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha)\}. \end{aligned} \quad (34)$$

Now note that

$$\begin{aligned} \prod_{i=1,2} \prod_{t=1}^{T_i} w_{i,D_{it}} &= \prod_{t=1}^{D^*} v_{0t}^{A_{1t}+A_{2t}} v_{1t}^{A_{1t}} v_{2t}^{A_{2t}} \\ &\times (1 - v_{0t} v_{1t})^{B_{1t}} (1 - v_{0t} v_{2t})^{B_{2t}}. \end{aligned} \quad (35)$$

Appendix B. Computational details

In one dimension the blocking structure of the Gibbs sampler described in Section 5 has been introduced by Papaspiliopoulos (2008) and Kalli et al. (2011) as an efficient alternative to the original algorithm of Walker (2007). A multidimensional slice sampling has been proposed by Hatjispyros et al. (2011) for a different dependent DP mixtures model. In this paper, we follow an alternative route and elaborate further on the Kalli et al. (2011) blocking strategy by proposing a nested structure for sampling stick variables. It should be noted that we borrow from Kalli et al. (2011) only the main blocking structure of the algorithm and leave for further research the use of auxiliary weights in the slicing scheme to increase the sampling efficiency.

B.1. General sampler

In what follows, if the kernels $\mathcal{K}_t$ depend on $[Y_{js} : j = 1, 2; s = 1, \dots, t-1]$, as in the case of the panel VAR model, we assume that $T_i = T$ for $i = 1, 2$. While, if the kernels $\mathcal{K}_t$ depend only on $[Y_{is} : s = 1, \dots, t-1]$, as in the case of the repeated measurements, we allow T_1 and T_2 to be different.

Define

$$A_{i,k} = \sum_{t=1}^{T_i} \mathbb{I}\{D_{i,t} = k\} = \text{card}(\mathcal{D}_{i,k}), \quad B_{i,k} = \sum_{t=1}^{T_i} \mathbb{I}\{D_{i,t} > k\}$$

and let

$$D^* = \max_{i=1,2} \max_{1 \leq t \leq T_i} D_{i,t}. \quad (36)$$

Updating $\tilde{\psi}$. The atoms $\tilde{\psi}$ given $[V, D, U, \tilde{\varphi}, Y]$ are conditionally independent and the full conditionals are:

$$\begin{aligned} P\{\tilde{\psi}_k \in d\psi_k | D, U, V, \tilde{\varphi}, Y\} &= P\{\tilde{\psi}_k \in d\psi_k | D, Y\} \\ &\propto G_0(d\psi_k) \prod_{t \in \mathcal{D}_{1,k}} \mathcal{K}_t(Y_{1,t} | \psi_{1k}, z_t) \prod_{t \in \mathcal{D}_{2,k}} \mathcal{K}_t(Y_{2,t} | \psi_{2k}, z_t); \end{aligned} \quad (37)$$

where $\psi_k = (\psi_{1k}, \psi_{2k})$. The strategy for sampling from this full conditional depends on the specific form of $\mathcal{K}_t$ and G_0 . In Appendices B.2 and B.3 we will discuss a possible strategy for Gaussian kernels used in Section 6.

Updating $[V, U, \tilde{\varphi}]$. In order to sample from the conditional distribution of $[V, U, \tilde{\varphi}]$ given $[D, \tilde{\psi}, Y]$ a further blocking is used:

- $[V^*, \tilde{\varphi}]$ given $[D, \tilde{\psi}, Y]$. By (34)–(35) it is easy to see that the joint conditional distribution of $[V^*, \tilde{\varphi}]$ given $[D, \tilde{\psi}, Y]$ is

$$\begin{aligned} P\{V^* \in dv^*, \tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha) | Y, \tilde{\psi}, D\} \\ = P\{V^* \in dv^*, \tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha) | D\} \end{aligned}$$

moreover, under (H1),

$$\begin{aligned} P\{V^* \in dv^*, \tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha) | D\} \\ \propto \prod_{k \in \mathcal{D}^*} \frac{Q_k(v_k | D, \varphi) dv_k \pi(d\theta_1, d\theta_2, d\alpha)}{B^2(\theta_1 + 1 - \alpha, \theta_2 + \alpha k) B(1 - \alpha, \theta_1)} \end{aligned} \quad (38)$$

with $v_k = (v_{0k}, v_{1k}, v_{2k})$ and

$$\begin{aligned} Q_k(v_k | D, \varphi) &= v_{0k}^{-\alpha + A_{1k} + A_{2k}} (1 - v_{0k})^{\theta_1 - 1} \\ &\times \prod_{i=1,2} v_{ik}^{A_{ik} + \theta_i - \alpha} (1 - v_{ik})^{\theta_2 + \alpha k - 1} (1 - v_{0k} v_{ik})^{B_{ik}}, \end{aligned}$$

while, under (H2),

$$\begin{aligned} P\{V^* \in dv^*, \tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha) | D\} \\ \propto \prod_{k \in \mathcal{D}^*} \frac{Q_k(v_k | D, \varphi) dv_k \pi(d\theta_1, d\theta_2, d\alpha)}{B(\theta_1 + 1 + \alpha(k-1), \theta_2) B(1 - \alpha, \theta_1 + \alpha k)} \end{aligned} \quad (39)$$

with $v_k = (v_{0k}, v_{1k})$ and

$$Q_k(v_k|D, \varphi) = v_{0k}^{A_{1k}+A_{2k}-\alpha} (1 - v_{0k})^{\theta_1+\alpha k+B_{2k}-1} \times v_{1k}^{\theta_1+\alpha(k-1)+A_{1k}} (1 - v_{1k})^{\theta_2-1} (1 - v_{0k}v_{1k})^{B_{1k}},$$

and $\pi(d\theta_1, d\theta_2, d\alpha) = P\{\tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha)\}$ is the prior on the stick-breaking parameters. To sample from (38)–(39), we iterate a two-step Metropolis–Hastings (M.–H.) within Gibbs with full conditionals

$$P\{V^* \in dv^*|\tilde{\varphi}, D\} \propto \prod_{k \in \mathcal{D}^*} Q_k(v_k|D, \tilde{\varphi}) dv_k \quad (40)$$

and

$$P\{\tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha)|V^*, D\} \propto \prod_{k \in \mathcal{D}^*} \frac{V_{0k}^{-\alpha} (1 - V_{0k})^{\theta_1}}{B(1 - \alpha, \theta_1)} \times \prod_{i=1,2} \frac{V_{ik}^{\theta_1-\alpha} (1 - V_{ik})^{\theta_2+\alpha k}}{B(\theta_1 + 1 - \alpha, \theta_2 + \alpha k)} \pi(d\theta_1, d\theta_2, d\alpha) \quad (41)$$

under (H1), and

$$P\{\tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha)|V^*, D\} \propto \prod_{k \in \mathcal{D}^*} \frac{V_{0k}^{-\alpha} (1 - V_{0k})^{\theta_1+\alpha k}}{B(1 - \alpha, \theta_1 + \alpha k)} \times \frac{V_{1k}^{\theta_1+\alpha(k-1)} (1 - V_{1k})^{\theta_2}}{B(\theta_1 + 1 + \alpha(k-1), \theta_2)} \pi(d\theta_1, d\theta_2, d\alpha) \quad (42)$$

under (H2). For the elements of V^* we consider a multivariate Gaussian random walk proposal with diagonal scale matrix and with scale parameter in the interval $[0.05, 0.3]$, in order to have acceptance rates between 0.3 and 0.5 for the elements of V^* . We generate samples from the full conditional of $\tilde{\varphi}$ by a M.–H. step. As the sampler depends on the prior specification we refer to [Appendices B.2 and B.3](#) for further details.

- V^{**} given $[D, V^*, \tilde{\psi}, \tilde{\varphi}, Y]$. The V_k (with $k \notin \mathcal{D}^*$) are conditionally independent given $[D, V^*, \tilde{\psi}, \tilde{\varphi}, Y]$ with $P\{V_k \in dv_k|\tilde{\varphi}, D, V^*\} \propto Q_k(v_k|D, \tilde{\varphi}) dv_k$ if $k \leq D^*$ and $P\{V_k \in dv_k|V^*, \tilde{\psi}, D, \tilde{\varphi}, Y\} = P\{V_k \in dv_k|\tilde{\varphi}\}$ if $k > D^*$. Note that if $k \notin \mathcal{D}^*$ and $k \leq D^*$, then $A_{i,k} = 0$ in the definition of $Q_k(v_k|D, \tilde{\varphi})$. In order to sample from $Q_k(v_k|D, \tilde{\varphi})$ the same M.–H. step, used for the full conditional in (40), is employed.
- U given $[V, D, \tilde{\psi}, \tilde{\varphi}, Y]$. The slice variables U are conditionally independent given $[V, D, \tilde{\psi}, \tilde{\varphi}, Y]$ with

$$P\{U_{i,t} \in du|V, Y, \tilde{\psi}, D\} = P\{U_{i,t} \in du|V, D\} = \frac{\mathbb{I}\{u \leq W_{i,D_{i,t}}\}}{W_{i,D_{i,t}}} du. \quad (43)$$

Updating D . The D 's are conditionally independent given $[V, U, \tilde{\psi}, \tilde{\varphi}, Y]$ with

$$P\{D_{i,t} = d|\tilde{\psi}, V, U, \tilde{\varphi}, Y\} \propto \mathcal{K}_t(Y_{i,t}|\tilde{\psi}_{id}, Z_t) \mathbb{I}\{U_{i,t} \leq W_{i,d}\}. \quad (44)$$

Here an important remark is in order. As in the slice sampling proposed in [Walker \(2007\)](#) and [Kalli et al. \(2011\)](#), the full conditional (44) samples, almost surely, from a finite number of terms. More precisely, $d > N_{i,t}^*$ ensures that $W_{i,d} < U_{i,t}$ where $N_{i,t}^*$ ($i = 1, 2; t = 1, \dots, T_i$) is the smallest integer such that

$$\sum_{k=1}^{N_{i,t}^*} W_{i,k} > 1 - U_{i,t}. \quad (45)$$

Note also that in the Block Gibbs Sampler described above in principle one needs to sample an infinite number of V_k and $\tilde{\psi}_k$. But in order to proceed with the chain it suffices to sample a finite number of V_k s to check condition (45) and the finite number of $\tilde{\psi}_k$ to be used in (44).

B.2. Full conditionals for $DPY(\varphi, G_0)$ mixtures of Gaussian of Section 6.1

For the sake of simplicity we will omit indicating the dependence of the full conditional on the hyperparameters. In order to sample from the full conditional $P\{\tilde{\psi}_k \in d\psi_k|V, D, Y, U\}$, for $k \geq 1$ we consider a Gibbs sampler with normal and inverse gamma full conditional distributions

$$\begin{aligned} P\{\tilde{\mu}_{ik} \in d\mu_{ik}|\tilde{\mu}_{0k}, \tilde{\sigma}_{ik}^2, \tilde{\sigma}_0^2, D, Y\} \\ \propto \exp\left\{-\frac{1}{2s_i^{-2}}\mu_{ik}^2\right\} \prod_{t \in \mathcal{D}_{i,k}} \exp\left\{-\frac{1}{2\tilde{\sigma}_{ik}^2\tilde{\sigma}_{0k}^2}\right. \\ \times (Y_{it} - (\tilde{\mu}_{0k} + \mu_{ik}))^2\left.\right\} d\mu_{ik} \\ \propto \exp\left\{-\frac{1}{2}\mu_{ik}^2 (s_i^2 + \tilde{\sigma}_{ik}^{-2}\tilde{\sigma}_{0k}^{-2}A_{i,k}) + \mu_{ik}\tilde{\sigma}_{ik}^{-2}\tilde{\sigma}_{0k}^{-2}\eta_{ik}^{(1)}\right\} d\mu_{ik} \\ \propto \mathcal{N}\left(\frac{\tilde{\sigma}_{ik}^{-2}\tilde{\sigma}_{0k}^{-2}\eta_{ik}^{(1)}}{s_i^2 + \tilde{\sigma}_{ik}^{-2}\tilde{\sigma}_{0k}^{-2}A_{i,k}}, \frac{1}{s_i^2 + \tilde{\sigma}_{ik}^{-2}\tilde{\sigma}_{0k}^{-2}A_{i,k}}\right) \end{aligned} \quad (46)$$

and

$$\begin{aligned} P\{\tilde{\sigma}_{ik}^2 \in d\sigma_{ik}^2|\tilde{\mu}_{0k}, \tilde{\mu}_{ik}, \tilde{\sigma}_{0k}^2, D, Y\} \\ \propto \exp\left\{-\frac{\lambda}{2}\sigma_{ik}^{-2}\right\} (\sigma_{ik}^{-2})^{\frac{1}{2}\lambda+1} \prod_{t \in \mathcal{D}_{i,k}} (\sigma_{ik}^{-2})^{1/2} \exp\left\{-\frac{1}{2\sigma_{ik}^2\tilde{\sigma}_{0k}^2}\right. \\ \times (Y_{it} - (\tilde{\mu}_{0k} + \tilde{\mu}_{ik}))^2\left.\right\} d\sigma_{ik}^2 \\ \propto \exp\left\{-\left(\frac{\lambda}{2} + \frac{1}{2\tilde{\sigma}_{0k}^2}\eta_{ik}^{(2)}\right)\sigma_{ik}^{-2}\right\} (\sigma_{ik}^{-2})^{\frac{1}{2}\lambda+\frac{1}{2}A_{i,k}+1} d\sigma_{ik}^2 \\ \propto \mathcal{IG}\left(\frac{\lambda}{2} + \frac{1}{2}A_{i,k}, \frac{\lambda}{2} + \frac{1}{2\tilde{\sigma}_{0k}^2}\eta_{ik}^{(2)}\right) \end{aligned} \quad (47)$$

$i = 1, 2$, for the component-specific part of the atoms, where $A_{i,k}$, $i = 1, 2$ have been defined in [Appendix B.1](#) and

$$\begin{aligned} \eta_{ik}^{(1)} &= \sum_{t \in \mathcal{D}_{i,k}} (Y_{it} - \tilde{\mu}_{0k}) \quad \text{and} \\ \eta_{ik}^{(2)} &= \sum_{t \in \mathcal{D}_{i,k}} (Y_{it} - (\tilde{\mu}_{0k} + \tilde{\mu}_{ik}))^2. \end{aligned} \quad (48)$$

The part of the atom that is common to all the components has full conditionals

$$\begin{aligned} P\{\tilde{\mu}_{0k} \in d\mu_{0k}|\tilde{\mu}_{1k}, \tilde{\mu}_{2k}, \tilde{\sigma}_{0k}^2, \tilde{\sigma}_{1k}^2, \tilde{\sigma}_{2k}^2, D, Y\} \\ \propto \exp\left\{-\frac{1}{2s_0^{-2}}\mu_{0k}^2\right\} \prod_{i=1,2} \prod_{t \in \mathcal{D}_{i,k}} \exp\left\{-\frac{1}{2\tilde{\sigma}_{ik}^2\tilde{\sigma}_{0k}^2}\right. \\ \times (Y_{it} - (\mu_{0k} + \tilde{\mu}_{ik}))^2\left.\right\} d\mu_{0k} \\ \propto \mathcal{N}\left(\frac{\eta_k^{(0)}}{s_0^2 + \sum_{i=1,2} \tilde{\sigma}_{0k}^{-2}\tilde{\sigma}_{ik}^{-2}A_{i,k}}, \frac{1}{s_0^2 + \sum_{i=1,2} \tilde{\sigma}_{0k}^{-2}\tilde{\sigma}_{ik}^{-2}A_{i,k}}\right) \end{aligned} \quad (49)$$

with

$$\eta_k^{(0)} = \sum_{i=1,2} \tilde{\sigma}_{0k}^{-2}\tilde{\sigma}_{ik}^{-2} \sum_{t \in \mathcal{D}_{i,k}} (Y_{it} - \tilde{\mu}_{ik}) \quad (50)$$

and

$$\begin{aligned}
 P\{\tilde{\sigma}_{0k}^2 \in d\sigma_{0k}^2 | \tilde{\sigma}_{1k}^2, \tilde{\sigma}_{2k}^2, \tilde{\mu}_0, \tilde{\mu}_1, \tilde{\mu}_2, D, Y\} \\
 \propto \exp\left\{-\frac{\varepsilon}{2}\sigma_{0k}^{-2}\right\} (\sigma_{0k}^{-2})^{\frac{\varepsilon}{2}+1} \prod_{i=1,2} \prod_{t \in \mathcal{D}_{i,k}} (\sigma_{0k}^2)^{-\frac{1}{2}} \\
 \times \exp\left\{-\frac{1}{2\tilde{\sigma}_{ik}^2\sigma_{0k}^2} (Y_{it} - (\tilde{\mu}_{0k} + \tilde{\mu}_{ik}))^2\right\} d\sigma_{0k}^2 \\
 \propto \mathcal{I}\mathcal{G}\left(\frac{\varepsilon}{2} + \frac{1}{2} \sum_{i=1,2} A_{ik}, \frac{\varepsilon}{2} + \frac{1}{2} \sum_{i=1,2} \frac{\eta_{ik}^{(2)}}{\tilde{\sigma}_{ik}^2}\right) \quad (51)
 \end{aligned}$$

respectively.

A sample from the conditional joint distribution of the precision parameters and the stick-breaking elements can be obtained following the blocking scheme described in Appendix B.1. Since we assume gamma prior distributions, $\mathcal{G}(\zeta_{11}, \zeta_{21})$ and $\mathcal{G}(\zeta_{12}, \zeta_{22})$ for θ_1 and θ_2 respectively, and standard uniform prior for α , equations (41)–(42) become

$$\begin{aligned}
 P\{\tilde{\varphi} \in (d\theta_1, d\theta_2, d\alpha) | V^*, D\} \propto \frac{1}{C(\varphi)} \theta_1^{\zeta_{11}-1} \theta_2^{\zeta_{12}-1} \exp\{-\theta_1 \bar{\zeta}_{21} \\
 - \theta_2 \bar{\zeta}_{22} - \alpha \bar{\zeta}_{32}\} d\theta_1 d\theta_2 d\alpha \quad (52)
 \end{aligned}$$

where

$$\begin{aligned}
 \bar{\zeta}_{21} &= \zeta_{21} - \sum_{k \in \mathcal{D}^*} \left(\log(1 - V_{0k}) + \sum_{i=1,2} \log V_{ik} \right) \\
 \bar{\zeta}_{22} &= \zeta_{22} - \sum_{k \in \mathcal{D}^*} \sum_{i=1,2} \log(1 - V_{ik}) \\
 \bar{\zeta}_{32} &= \sum_{k \in \mathcal{D}^*} \left(\log V_{0k} + \sum_{i=1,2} (\log V_{ik} - k \log(1 - V_{ik})) \right) \\
 C(\varphi) &= \prod_{k \in \mathcal{D}^*} B(1 - \alpha, \theta_1) B^2(\theta_1 + 1 - \alpha, \theta_2 + \alpha k) \quad (53)
 \end{aligned}$$

under (H1), and

$$\begin{aligned}
 \bar{\zeta}_{21} &= \zeta_{21} - \sum_{k \in \mathcal{D}^*} (\log(1 - V_{0k}) + \log(V_{1k})) \\
 \bar{\zeta}_{22} &= \zeta_{22} - \sum_{k \in \mathcal{D}^*} \log(1 - V_{1k}) \\
 \bar{\zeta}_{32} &= \sum_{k \in \mathcal{D}^*} (\log(V_{0k}) - k \log(1 - V_{0k}) - (k - 1) \log(V_{1k})) \\
 C(\varphi) &= \prod_{k \in \mathcal{D}^*} B(1 - \alpha, \theta_1 + \alpha k) B(\theta_1 + 1 + \alpha(k - 1), \theta_2) \quad (54)
 \end{aligned}$$

under (H2).

We simulate from the full conditional by a M.–H. step. We considered a Gaussian random walk proposal for the transformed parameter vector $\xi = (\xi_1, \xi_2, \xi_3) = g(\varphi)$ with $g(\varphi) = (\log(\theta_1), \log(\theta_2), \log(\alpha/(1 - \alpha)))$. At the j th iteration, given $\varphi^{(j-1)}$, we simulate

$$\xi^{(*)} \sim \mathcal{N}_3(\xi^{(j-1)}, \kappa^2 I_3) \quad (55)$$

where $\xi^{(j-1)} = g(\varphi^{(j-1)})$ and κ^2 represents a scale parameter of random walk. The proposal $\varphi^{(*)} = g^{-1}(\xi^{(*)})$, with g^{-1} inverse transform of g , is accepted with probability

$$\min\left\{1, \frac{P\{\varphi^{(*)} | V^*, D\} |\nabla g^{-1}(\varphi^{(*)})|}{P\{\varphi^{(j-1)} | V^*, D\} |\nabla g^{-1}(\varphi^{(j-1)})|}\right\} \quad (56)$$

where ∇g^{-1} denotes the Jacobian of g^{-1} , that is a matrix with main diagonal

$$\left(\exp(\xi_1), \exp(\xi_2), \frac{\exp(\xi_3)}{(1 + \exp(\xi_3))^2} \right) \quad (57)$$

and null elements out of the main diagonal and $|\nabla g^{-1}|$ its determinant. We set the scale parameter κ in order to have acceptance rates close to 0.5.

B.3. Full conditionals for DPY(φ, G_0) mixtures of VAR of Section 6.2

In order to sample from the full-conditional $P\{\tilde{\psi}_k \in d\psi_k | V, D, Y, U\}$, for $k \geq 1$, we use Gibbs sampling. The hierarchical structure of the base measure allows us to write the following Gibbs sampling updating scheme for the elements of $\tilde{\psi}_k$.

The full conditional of the country-specific VAR coefficients for the i th unit is

$$\begin{aligned}
 P\{\tilde{\phi}_{ik} \in d\phi_{ik} | \tilde{\phi}_{0k}, \tilde{\Sigma}_{ik}, \tilde{\sigma}_{0k}^2, \tilde{\gamma}_{ik}^2, D, Y\} \\
 \propto \exp\left\{-\frac{1}{2} \text{tr}\left(\sum_{t \in \mathcal{D}_{i,k}} (E_{0ikt} - (I_2 \otimes X_t') \phi_{ik})' \tilde{\Sigma}_{ik}^{-1} \tilde{\sigma}_{0k}^{-2} \right. \right. \\
 \times \left. \left. (E_{0ikt} - (I_2 \otimes X_t') \phi_{ik}) + \phi_{ik}' \tilde{\gamma}_{ik}^{-1} \phi_{ik}\right)\right\} d\phi_{ik} \\
 \propto \exp\left\{-\frac{1}{2} \text{tr}\left(\phi_{ik}' \left(\sum_{t \in \mathcal{D}_{i,k}} (I_2 \otimes X_t) \tilde{\Sigma}_{ik}^{-1} \tilde{\sigma}_{0k}^{-2} (I_2 \otimes X_t') \right. \right. \right. \\
 \left. \left. + \tilde{\gamma}_{ik}^{-1}\right) \phi_{ik} - 2 \sum_{t \in \mathcal{D}_{i,k}} \phi_{ik}' ((I_2 \otimes X_t) \tilde{\Sigma}_{ik}^{-1} \tilde{\sigma}_{0k}^{-2} E_{0ikt})\right\} d\phi_{ik} \\
 \propto \mathcal{N}_m(m_{ik}, M_{ik})
 \end{aligned}$$

where $E_{0ikt} = Y_{it} - (I_2 \otimes X_t') \tilde{\phi}_{0k}$, $M_{ik}^{-1} = \tilde{\sigma}_{0k}^{-2} \sum_{t \in \mathcal{D}_{i,k}} (I_2 \otimes X_t) \tilde{\Sigma}_{ik}^{-1} (I_2 \otimes X_t') + \tilde{\gamma}_{ik}^{-1}$ and $m_{ik} = M_{ik} \tilde{\sigma}_{0k}^{-2} \sum_{t \in \mathcal{D}_{i,k}} (I_2 \otimes X_t) \tilde{\Sigma}_{ik}^{-1} E_{0ikt}$.

The country-specific variance matrix of the VAR equations for the i th unit has full conditional

$$\begin{aligned}
 P\{\tilde{\Sigma}_{ik}^{-1} \in d\Sigma_{ik}^{-1} | \tilde{\phi}_{ik}, \tilde{\phi}_{0k}, \tilde{\sigma}_{0k}^2, \tilde{\gamma}_{ik}^2, D, Y\} \\
 \propto \exp\left\{-\frac{1}{2} \sum_{t \in \mathcal{D}_{i,k}} \text{tr}(E_{ikt}' \Sigma_{ik}^{-1} \tilde{\sigma}_{0k}^{-2} E_{ikt})\right\} |\Sigma_{ik}|^{-\frac{A_{ik}}{2}} |\Sigma_{ik}^{-1}|^{\frac{\lambda-3}{2}} \\
 \times \exp\left\{-\frac{1}{2} \text{tr}(\Lambda^{-1} \Sigma_{ik}^{-1})\right\} d\Sigma_{ik} \\
 \propto \exp\left\{-\frac{1}{2} \text{tr}\left(\left(\Lambda^{-1} + \tilde{\sigma}_{0k}^{-2} \sum_{t \in \mathcal{D}_{i,k}} E_{ikt} E_{ikt}'\right) \Sigma_{ik}^{-1}\right)\right\} \\
 \times |\Sigma_{ik}^{-1}|^{\frac{A_{ik} + \lambda - 3}{2}} d\Sigma_{ik} \\
 \propto \mathcal{W}_2(\lambda + A_{ik}, \Lambda_{i,T})
 \end{aligned}$$

where $A_{i,k} = \text{card}(\mathcal{D}_{i,k})$, $E_{ikt} = Y_{it} - (I_2 \otimes X_t')(\tilde{\phi}_{0k} + \tilde{\phi}_{ik})$ and $\Lambda_{i,T}^{-1} = \Lambda^{-1} + \tilde{\sigma}_{0k}^{-2} \sum_{t \in \mathcal{D}_{i,k}} E_{ikt} E_{ikt}'$.

The full conditionals of the common components $\tilde{\phi}_{0k}$ and σ_{0k}^{-2} are functions of the unit-specific components. More specifically $\tilde{\phi}_{0k}$ has full conditional

$$\begin{aligned}
 P\{\tilde{\phi}_{0k} \in d\phi_{0k} | \tilde{\phi}_{ik}, \tilde{\Sigma}_{ik}, \tilde{\sigma}_{0k}^2, \tilde{\gamma}_{ik}^2, D, Y\} \\
 \propto \exp\left\{-\frac{1}{2} \text{tr}\left(\phi_{0k}' \left(\sum_{i=1,2} \sum_{t \in \mathcal{D}_{i,k}} (I_2 \otimes X_t) \right. \right. \right. \\
 \times \left. \left. \tilde{\Sigma}_{ik}^{-1} \tilde{\sigma}_{0k}^{-2} (I_2 \otimes X_t') + \tilde{\gamma}_0^{-1}\right) \phi_{0k} \right. \\
 \left. - 2 \sum_{i=1,2} \sum_{t \in \mathcal{D}_{i,k}} \phi_{0k}' ((I_2 \otimes X_t) \tilde{\Sigma}_{ik}^{-1} \tilde{\sigma}_{0k}^{-2} E_{1ikt})\right\} d\phi_{0k} \\
 \propto \mathcal{N}_m(m_{0k}, M_{0k})
 \end{aligned}$$

Table 4

Posterior means ($\hat{\theta}_1, \hat{\theta}_2, \hat{\alpha}$), 95% credibility intervals (CI) and posterior standard deviation (Std) of the DPY process parameters $\varphi = (\theta_1, \theta_2, \alpha)$, the posterior and the prior expected number of components for the first, $k_1 = \mathbb{E}(K_T|Y)$ and $\bar{k}_1 = \mathbb{E}(K_T)$, and the second, $k_2 = \mathbb{E}(K_T|Y)$ and $\bar{k}_2 = \mathbb{E}(K_T)$, group of observations.

Prior	Parameter estimates									Clusters			
	$\hat{\theta}_1$	CI	Std	$\hat{\theta}_2$	CI	Std	$\hat{\alpha}$	CI	Std	$\bar{k}_1$	$\bar{k}_2$	k_1	k_2
Mix1													
$\beta_1 - DD(\varphi, G_0)$	0.07	[0.01, 0.25]	0.06	0.08	[0.01, 0.29]	0.07				3.02	3.77	8.39	8.39
$\beta_2 - DD(\varphi, G_0)$	0.01	[0.01, 0.46]	0.12	0.16	[0.01, 0.59]	0.44				3.16	3.04	8.39	5.18
$\beta_1 - DPY(\varphi, G_0)$	0.24	[0.01, 0.59]	0.15	0.34	[0.11, 0.82]	0.23	0.04	[0.01, 0.17]	0.04	3.28	3.30	9.30	9.30
$\beta_2 - DPY(\varphi, G_0)$	0.33	[0.04, 0.97]	0.31	0.23	[0.03, 0.73]	0.17	0.05	[0.01, 0.23]	0.05	3.20	3.09	9.30	5.87
Mix2													
$\beta_1 - DD(\varphi, G_0)$	0.09	[0.01, 0.46]	0.14	0.03	[0.01, 0.13]	0.0348				4.76	2.08	8.39	8.39
$\beta_2 - DD(\varphi, G_0)$	0.05	[0.01, 0.16]	0.04	0.09	[0.01, 0.33]	0.09				4.03	3.98	8.39	5.18
$\beta_1 - DPY(\varphi, G_0)$	0.04	[0.01, 0.19]	0.04	0.58	[0.07, 1.36]	0.40	0.02	[0.01, 0.07]	0.01	4.55	3.05	9.30	9.30
$\beta_2 - DPY(\varphi, G_0)$	0.61	[0.05, 1.35]	0.42	0.34	[0.08, 0.98]	0.23	0.05	[0.01, 0.19]	0.05	4.14	4.06	9.30	5.87
Mix3													
$\beta_1 - DD(\varphi, G_0)$	0.08	[0.01, 0.32]	0.09	0.08	[0.02, 0.27]	0.06				3.93	4.03	8.39	8.39
$\beta_2 - DD(\varphi, G_0)$	0.12	[0.01, 0.40]	0.10	0.10	[0.01, 0.45]	0.12				3.16	3.03	8.39	5.18
$\beta_1 - DPY(\varphi, G_0)$	0.19	[0.03, 0.89]	0.23	0.65	[0.11, 2.10]	0.52	0.03	[0.01, 0.11]	0.03	3.54	3.57	9.30	9.30
$\beta_2 - DPY(\varphi, G_0)$	0.27	[0.01, 0.92]	0.24	0.48	[0.08, 1.44]	0.40	0.05	[0.01, 0.19]	0.04	6.86	3.82	9.30	5.87
Mix4													
$\beta_1 - DD(\varphi, G_0)$	0.11	[0.01, 0.39]	0.09	0.05	[0.01, 0.17]	0.04				3.05	3.03	8.39	8.39
$\beta_2 - DD(\varphi, G_0)$	0.07	[0.01, 0.35]	0.23	0.05	[0.01, 0.22]	0.13				4.04	3.13	8.39	5.18
$\beta_1 - DPY(\varphi, G_0)$	0.11	[0.02, 0.25]	0.07	1.18	[0.49, 2.43]	0.58	0.01	[0.01, 0.05]	0.01	6.49	3.98	9.30	9.30
$\beta_2 - DPY(\varphi, G_0)$	0.09	[0.01, 0.25]	0.06	1.21	[0.46, 2.40]	0.49	0.02	[0.01, 0.06]	0.01	6.97	3.12	9.30	5.87

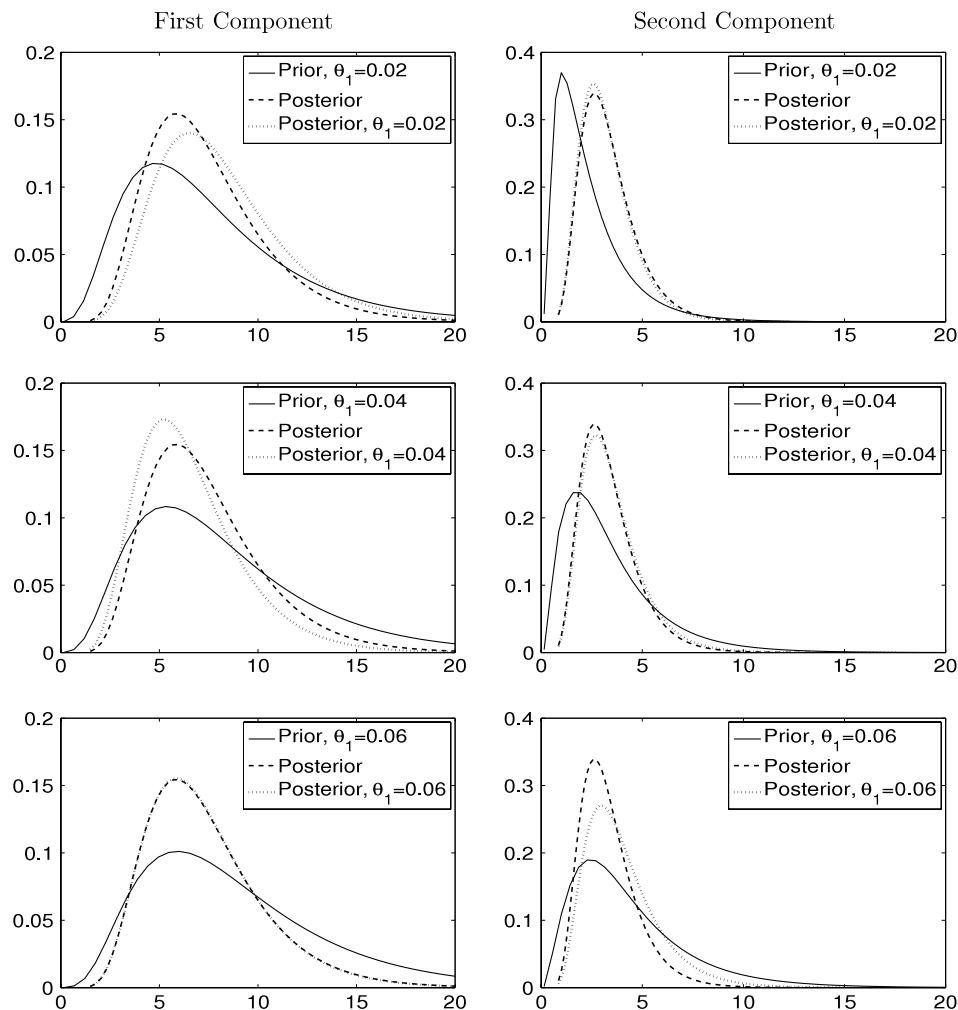


Fig. 13. Original posterior distribution (dashed lines) and prior (solid lines) and posterior (dotted lines) distributions of the number of clusters, K_T for $\theta_2 = 1.21$, $\alpha = 0.02$, $T = 100$ and varying θ_1 .

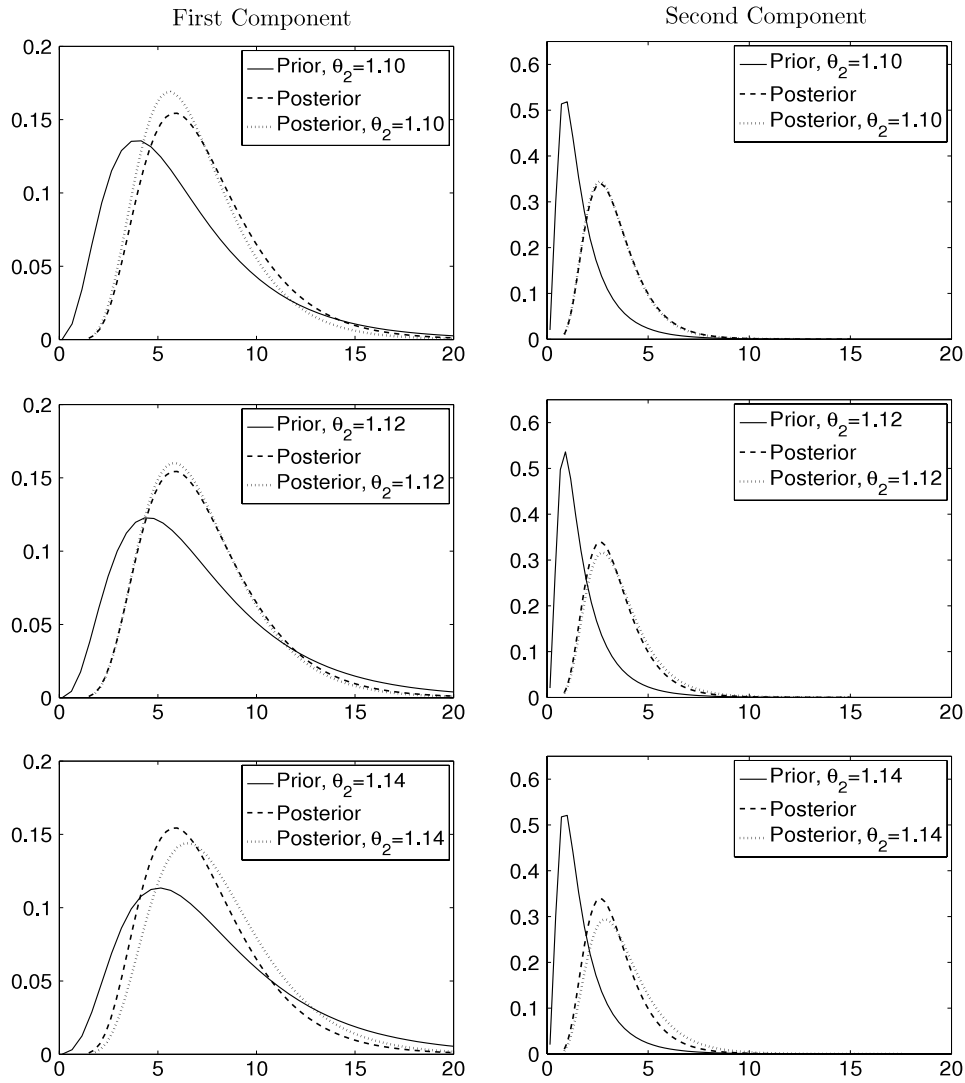


Fig. 14. Original posterior distribution (dashed lines) and prior (solid lines) and posterior (dotted lines) distributions of the number of clusters, K_T for $\theta_1 = 0.09$, $\alpha = 0.02$, $T = 100$ and varying θ_2 .

where $E_{1ikt} = Y_{it} - (I_2 \otimes X'_t)\tilde{\phi}_{ik}$, $M_{0k}^{-1} = \tilde{\sigma}_{0k}^{-2} \sum_{i=1,2} \sum_{t \in \mathcal{D}_{i,k}} (I_2 \otimes X_t) \tilde{\Sigma}_{ik}^{-1} (I_2 \otimes X'_t) + \tilde{\gamma}_0^{-1}$ and $m_{0k} = M_{0k} \tilde{\sigma}_{0k}^{-2} \sum_{i=1,2} \sum_{t \in \mathcal{D}_{i,k}} (I_2 \otimes X_t) \tilde{\Sigma}_{ik}^{-1} E_{1ikt}$.

Finally the common factor of the variances, σ_{0k}^{-2} , has full conditional

$$\begin{aligned} P\{\tilde{\sigma}_{0k}^{-2} \in d\sigma_{0k}^{-2} | \tilde{\phi}_{ik}, \tilde{\Sigma}_{ik}, \tilde{\phi}_{0k}, \tilde{\gamma}_{ik}^2, D, Y\} \\ \propto \exp \left\{ -\frac{1}{2\sigma_{0k}^2} \sum_{i=1,2} \sum_{t \in \mathcal{D}_{i,k}} (E'_{ikt} \tilde{\Sigma}_{ik}^{-1} E_{ikt}) \right\} (\sigma_{0k}^{-2})^{\frac{A_{1,k} + A_{2,k}}{2}} (\sigma_{0k}^{-2})^{\frac{\varepsilon-1}{2}} \\ \times \exp \left\{ -\frac{\varepsilon}{2\sigma_{0k}^2} \right\} d\sigma_{0k}^{-2} \\ \propto \mathcal{G}a \left(\left(\varepsilon + \sum_{i=1,2} A_{ik} \right) / 2, \varepsilon_T / 2 \right) \end{aligned}$$

where $\varepsilon_T = \varepsilon + \sum_{i=1,2} \sum_{t \in \mathcal{D}_{i,k}} E'_{ikt} \tilde{\Sigma}_{ik}^{-1} E_{ikt}$ and

$$\begin{aligned} P\{\tilde{\tau}_k^2 \in d\tau_k^2 | \tilde{\phi}_{ik}, \tilde{\Sigma}_{ik}, \tilde{\phi}_{0k}, \tilde{\sigma}_{0k}^{-2}, D, Y\} \\ \propto \exp \left\{ -\frac{1}{2\tau_k^2} \text{tr} \left(\sum_{i=1,2} \tilde{\phi}'_{ik} \tilde{\phi}_{ik} \right) \right\} \tau_k^{-4} \tau_k^{-\nu-1/2} \exp\{-\nu/2\tau_k^{-2}\} \end{aligned}$$

$$\begin{aligned} \propto \exp \left\{ -\frac{1}{2} \left(\nu + \text{tr} \left(\sum_{i=1,2} \tilde{\phi}'_{ik} \tilde{\phi}_{ik} \right) \right) \tau_k^{-2} \right\} \tau_k^{-2(\nu/2+1)} \\ \propto \mathcal{I} \mathcal{G} \left(\nu/2 + 2, \nu/2 + \text{tr} \left(\sum_{i=1,2} \tilde{\phi}'_{ik} \tilde{\phi}_{ik} \right) / 2 \right). \end{aligned}$$

Appendix C. Sensitivity analysis

C.1. Estimation details

Table 4 reports the posterior mean, standard deviation and 95% credibility interval for the DPY parameters. Note that the posterior standard deviations are in the $[0, 0.6]$ interval in all the experiments, while the standard deviation of the prior distribution is 10. This shows that data are informative for the parameter φ . The last four columns of Table 4 report the posterior and the prior expected number of clusters for the first and the second component of the DPY prior is denoted by $\bar{k}_1$ and $\bar{k}_2$ respectively. It has been evaluated analytically (see Section 6) using the prior mean of φ . $\bar{k}_1$ and $\bar{k}_2$ are generally larger than the posterior number of clusters, $\bar{k}_i$, $i = 1, 2$, and are comparable in DD and DPY, within each assumptions (H1) and (H2).

Table 5

Posterior and prior means and standard deviations (in parenthesis) of the number of clusters for the first and second components of the DPY process for the dataset Mix4. The sample size is $T = 100$ and the parameter settings are: $\theta_1 = 0.09$, $\theta_2 = 1.21$, varying α (a); $\theta_2 = 1.21$ and $\alpha = 0.02$, varying θ_1 (b); $\theta_1 = 0.09$ and $\alpha = 0.02$, varying θ_2 (c).

(a)				
α	$\bar{k}_1$	$\bar{k}_2$	k_1	k_2
0.1	6.59 (0.95)	3.10 (0.32)	7.93 (2.81)	2.81 (1.36)
0.2	6.73 (1.25)	3.20 (0.46)	10.08 (3.73)	3.51 (2.33)
0.3	7.26 (1.32)	3.89 (0.68)	13.04 (4.94)	5.26 (3.69)
(b)				
θ_1	$\bar{k}_1$	$\bar{k}_2$	k_1	k_2
0.2	7.65 (1.10)	3.02 (0.13)	6.89 (2.25)	2.12 (1.06)
0.4	6.57 (0.87)	3.24 (0.46)	7.56 (2.35)	3.04 (1.42)
0.6	6.90 (0.96)	3.72 (0.78)	8.18 (2.47)	3.89 (1.67)
(c)				
θ_2	$\bar{k}_1$	$\bar{k}_2$	k_1	k_2
1.0	6.48 (0.63)	3.08 (0.27)	6.47 (2.61)	1.58 (1.02)
1.2	6.79 (0.78)	3.29 (0.48)	6.48 (2.21)	1.58 (1.02)
1.4	7.37 (0.81)	3.15 (0.38)	7.14 (2.83)	1.58 (1.02)

C.2. Sensitivity analysis with respect to θ_1 and θ_2

We check the robustness of the results with respect to changes in the values of θ_1 and θ_2 , when setting $\alpha = 0.02$. The prior and posterior distributions of the number of clusters, are given in Figs. 13–14 and Table 5. The results show that the prior dispersion is not strongly affected by changes in θ_1 and θ_2 , and that an increase in value of these parameters leads to a shift to the right of the distribution. In all our experiments, we found that the posterior expected number of clusters is robust to changes in the prior specification. As a side remark, we note that the proposed changes in the concentration parameters θ_i , $i = 1, 2$, are not altering significantly the computing time of the MCMC procedure. In all the experiments of the second and the third set, 10,000 MCMC iterations take about 15 min.

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